ABSTRACT

This project addresses the growing need for low-cost, flexible, and easily manufacturable electronic devices by exploring the use of organic semiconductor materials. Conventional silicon-based electronics, while highly efficient, involve complex fabrication processes, high costs, and limited flexibility. These limitations restrict their application in emerging fields such as wearable electronics, flexible displays, and large-area devices. To overcome these challenges, this project investigates the use of Polyaniline (PANI), a conductive polymer, as an alternative material for device fabrication.

The primary objective of this work is to design, fabricate, and analyze three fundamental electronic devices using PANI: a Metal–Semiconductor–Metal (MSM) structure, a MOS capacitor, and a Thin Film Transistor (TFT). The methodology involves a series of fabrication steps including substrate cleaning, spin coating of PANI thin films, metal deposition (Aluminum) for electrodes, dielectric layer formation using Al_2O_3, and photolithography for precise patterning. The project was carried out in two phases, with MSM and MOS capacitor devices fabricated in the first phase, followed by TFT fabrication and optimization in the second phase.

The MSM device was utilized to study the current–voltage (I–V) characteristics of PANI, providing insight into its conductivity and charge transport behavior. The MOS capacitor enabled the analysis of capacitance behavior and charge storage properties, highlighting the importance of dielectric-semiconductor interfaces. The Thin Film Transistor demonstrated switching behavior, confirming the applicability of PANI in active electronic devices. Although the devices showed functional behavior, certain challenges such as non-uniform thin films, environmental sensitivity, and limited performance compared to conventional semiconductors were observed.

From a societal perspective, this project contributes to the development of affordable and sustainable electronics. The use of solution-processable materials and simple fabrication techniques reduces manufacturing costs and energy consumption. Additionally, organic electronics enable the development of flexible and lightweight devices, which are highly suitable for applications in healthcare monitoring, wearable technology, and portable systems.

In conclusion, this project successfully demonstrates the feasibility of using Polyaniline for fabricating basic electronic devices and highlights its potential for future applications in low-cost and flexible electronics, while also identifying key areas for improvement in organic semiconductor technology.

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List of Abbreviations

Abbreviation	Full Form
PANI	Polyaniline
MSM	Metal–Semiconductor–Metal
MOS	Metal–Oxide–Semiconductor
TFT	Thin Film Transistor
Al	Aluminum
Al_2O_3	Aluminum Oxide
I-V	Current–Voltage
C-V	Capacitance–Voltage
RF	Radio Frequency
DC	Direct Current
UV	Ultraviolet
IPA	Isopropyl Alcohol
DI	Deionized
NMP	N-Methyl-2-pyrrolidone
HOMO	Highest Occupied Molecular Orbital
LUMO	Lowest Unoccupied Molecular Orbital
FET	Field-Effect Transistor
OECT	Organic Electrochemical Transistor
PEDOT:PSS	Poly(3,4-ethylenedioxythiophene) Polystyrene Sulfonate
APS	Ammonium Persulfate
IDE	Interdigitated Electrode

Chapter 1: Introduction

1.1 Background

The rapid advancement of modern electronics has been primarily driven by silicon-based semiconductor technology. While silicon devices offer high performance and reliability, they involve complex fabrication processes, high production costs, and limited mechanical flexibility. With the increasing demand for low-cost, lightweight, and flexible electronic systems, there is a growing need to explore alternative materials and device architectures.

Organic electronics has emerged as a promising field that utilizes carbon-based materials for device fabrication. Among various organic materials, Polyaniline (PANI) has gained significant attention due to its excellent electrical conductivity, environmental stability, ease of synthesis, and cost-effectiveness. Additionally, PANI can be processed using simple techniques such as spin coating and drop casting, making it suitable for scalable and flexible device fabrication.

The motivation behind this project lies in investigating the feasibility of using PANI as an active material in fundamental electronic devices. By fabricating and analyzing three essential device structures—Metal–Semiconductor–Metal (MSM), MOS capacitor, and Thin Film Transistor (TFT)—this work aims to understand the electrical behavior and practical applicability of organic semiconductors.

The need for this study is further driven by the growing interest in flexible electronics, wearable devices, and large-area electronics, where traditional silicon-based technologies face limitations. Organic materials like PANI provide a pathway toward developing such next-generation devices with simpler fabrication processes and reduced manufacturing costs.

Thus, this project focuses on bridging the gap between conventional semiconductor technology and emerging organic electronics by demonstrating the fabrication and functionality of PANI-based devices.

1.2 Problem Definition and Objectives

Problem Definition

Conventional silicon-based electronic devices, while highly efficient, involve complex fabrication processes, high costs, and limited flexibility. These limitations make them less suitable for emerging applications such as flexible electronics, wearable systems, and low-cost large-area devices. Organic semiconductors like Polyaniline (PANI) offer a promising alternative due to their solution-processability and mechanical flexibility; however, their integration into reliable electronic devices and understanding of their electrical behavior remain key challenges.

The specific technical challenge addressed in this project is the fabrication and evaluation of fundamental electronic devices using PANI as the active material, while ensuring proper layer formation, interface quality, and functional device operation.

Objectives

• To fabricate a Metal–Semiconductor–Metal (MSM) device using PANI and analyze its current–voltage (I–V) characteristics.

• To develop a MOS capacitor structure using PANI and Al_2O_3 dielectric, and study its capacitance behavior.

• To fabricate a Thin Film Transistor (TFT) using PANI and evaluate its switching characteristics.

• To optimize fabrication steps such as spin coating, metal deposition, and lithography for organic device integration.

1.3 Social & Environmental Relevance

The growing demand for sustainable and affordable electronic technologies has increased the importance of exploring alternative materials and fabrication methods. Traditional semiconductor manufacturing relies on energy-intensive processes, expensive infrastructure, and non-renewable materials, which contribute to environmental concerns and high production costs. In contrast, organic electronics based on materials like Polyaniline (PANI) offer a more eco-friendly and economically viable approach.

From a social perspective, the use of low-cost and easily processable materials such as PANI can enable the development of affordable electronic devices, making technology more accessible, especially in developing regions. Applications such as low-cost sensors, wearable health monitoring systems, and flexible electronic devices can significantly improve quality of life and support advancements in healthcare, education, and communication.

Environmentally, PANI-based devices can be fabricated using solution-based techniques like spin coating and printing, which consume less energy compared to conventional semiconductor fabrication processes. Additionally, the potential for flexible and lightweight devices reduces material usage and supports the development of portable, energy-efficient systems. Organic materials also open possibilities for greener disposal and reduced electronic waste, although further research is needed in this area.

Overall, this project contributes to the broader goal of developing sustainable, cost-effective, and flexible electronic technologies by demonstrating the feasibility of organic semiconductor devices.

1.4 Project Plan

1.4.1 Work Breakdown Structure

The project was divided into two major phases corresponding to academic evaluation stages:

Phase 1 (Major Project – I):
- Literature review on organic electronics and Polyaniline (PANI)
- Procurement of materials and lab familiarization
- Fabrication of Metal–Semiconductor–Metal (MSM) device
- Fabrication of MOS capacitor structure
- Initial testing and verification of fabricated devices

Phase 2 (Major Project – II):
- Process optimization based on Phase 1 observations
- Fabrication of Thin Film Transistor (TFT) using PANI
- Improvement in lithography and layer deposition techniques
- Electrical characterization and comparative analysis of all devices
- Documentation and report preparation

1.4.2 Timeline

The project was carried out over two semesters:

Semester 7 (Major I):
• Weeks 1–3: Literature review and understanding fabrication techniques
• Weeks 4–6: Substrate preparation and practice of spin coating
• Weeks 7–10: Fabrication of MSM device
• Weeks 11–14: Fabrication of MOS capacitor
• Weeks 15–16: Initial testing and evaluation

Semester 8 (Major II):
• Weeks 1–3: Review and process optimization
• Weeks 4–8: TFT fabrication (lithography + deposition)
• Weeks 9–12: Device testing and troubleshooting
• Weeks 13–15: Data analysis and comparison
• Weeks 16: Final report and presentation preparation

1.4.3 Roles and Responsibilities

Both team members actively participated in all stages of the project. The work was carried out collaboratively with continuous coordination and supervision during laboratory processes.

• Both members were present during all fabrication steps, ensuring accuracy and consistency
• Joint monitoring of critical processes such as spin coating, lithography, and deposition
• Equal involvement in device fabrication, testing, and troubleshooting
• Shared responsibility for data collection, analysis, and documentation
• Collaborative preparation of final report and presentation

This collaborative approach ensured better precision in fabrication and minimized experimental errors.

1.4.4 Challenges and Risk Management

The project involved several practical challenges associated with organic device fabrication and laboratory processes:

Challenges Faced:

• Difficulty in achieving uniform PANI thin films during spin coating
• Sensitivity of organic materials to environmental conditions such as humidity and temperature
• Alignment issues during lithography affecting electrode patterns
• Possible defects in thin films such as pinholes or non-uniform thickness
• Adhesion issues between different material layers (Al, PANI, Al_2O_3)
• Limited availability and precision of fabrication equipment

Risk Management Strategies:

• Careful cleaning of substrates to improve film uniformity and adhesion
• Optimization of spin coating parameters (speed, time, solution concentration)
• Performing fabrication in controlled lab conditions as much as possible
• Repeating critical steps like coating and lithography to minimize defects
• Close monitoring during each fabrication stage to detect errors early
• Maintaining proper handling and storage of chemicals and materials
• Continuous consultation with lab supervisors and faculty for troubleshooting

By addressing these challenges systematically, the team ensured successful fabrication and improved reliability of the devices.

1.5 Scope and Limitations

Scope

This project focuses on the fabrication and basic characterization of organic electronic devices using Polyaniline (PANI) as the active material. The scope of the work includes:

• Design and fabrication of three fundamental devices: Metal–Semiconductor–Metal (MSM), MOS capacitor, and Thin Film Transistor (TFT)

• Utilization of simple and cost-effective fabrication techniques such as spin coating, metal deposition, and photolithography

• Study of basic electrical behavior including current–voltage (I–V) characteristics and qualitative device response

• Understanding the feasibility of integrating organic materials into conventional device structures

• Gaining practical exposure to microfabrication processes and organic semiconductor handling

The project provides a foundational understanding of organic electronics and demonstrates the potential of PANI for low-cost and flexible device applications.

Limitations

Despite successful fabrication, the project has certain limitations:

• Lack of advanced characterization tools for precise measurement of electrical parameters (e.g., mobility, threshold voltage accuracy)

• Possible non-uniformity in PANI thin films due to limitations in fabrication control

• Environmental sensitivity of organic materials affecting device stability and repeatability

• Limited optimization of device dimensions and material thicknesses

• Absence of high-end cleanroom facilities, leading to possible contamination and defects

• Performance of fabricated devices may not match that of commercial silicon-based devices

These limitations highlight areas for future improvement and further research in organic device fabrication and optimization

Chapter 2: Literature Survey

2.1 Critical Review

Organic electronics has emerged as a promising alternative to conventional silicon-based technology due to its advantages such as low-cost fabrication, mechanical flexibility, and solution processability. Among various conducting polymers, Polyaniline (PANI) has attracted significant attention because of its tunable electrical conductivity, environmental stability, and ease of synthesis [1], [2]. Several researchers have explored the application of PANI in electronic devices, particularly in thin film transistors, conductive films, and sensing systems.

Early work demonstrated the feasibility of using PANI as an active semiconductor material in field-effect transistor (FET) structures. These studies showed that PANI-based thin films can exhibit transistor-like switching behavior when used in appropriate device configurations [8], [9]. However, the performance of these devices was limited by low charge carrier mobility and strong dependence on fabrication conditions. Additionally, achieving uniform thin films was identified as a critical challenge affecting device consistency and repeatability.

Further developments in organic transistor technology led to the exploration of organic electrochemical transistors (OECTs), where PANI and similar conducting polymers were used due to their ability to undergo reversible doping and dedoping. These devices demonstrated improved sensitivity and conductivity modulation, making them suitable for sensing applications [4]. Despite these advantages, issues related to long-term stability, non-linear response, and environmental sensitivity remain significant challenges.

Research has also focused on improving the structural and electrical properties of PANI thin films. Advanced material engineering techniques have been employed to enhance conductivity and device performance. For instance, the development of crystalline quasi-two-dimensional PANI films resulted in significantly improved conductivity and electrical characteristics [6]. Such improvements highlight the importance of material structure and morphology in determining device behavior. However, these approaches often require sophisticated fabrication techniques and controlled environments, limiting their accessibility.

In addition, several studies have emphasized the role of fabrication techniques in determining device performance. Solution processing methods such as spin coating have been widely used due to their simplicity and cost-effectiveness [5]. These methods enable the fabrication of large-area thin films suitable for flexible electronics. However, maintaining uniformity and minimizing defects in such films remains a challenge, especially in non-controlled environments.

Organic thin film transistors have also been extensively studied for sensing and flexible electronics applications. These devices offer advantages such as low power consumption and compatibility with flexible substrates [3], [8]. However, their performance is often inferior to conventional silicon-based devices, particularly in terms of mobility, switching speed, and long-term reliability.

Overall, the literature indicates that while Polyaniline is a highly promising material for organic electronic devices, several challenges remain unresolved. These include improving charge carrier mobility, achieving uniform thin films, enhancing environmental stability, and simplifying fabrication processes. Furthermore, most existing studies focus on individual device structures rather than exploring multiple device types using the same material system. This highlights the need for comprehensive experimental studies that investigate the behavior of PANI across different electronic device configurations.

2.2 Comparison of papers

Ref. No.	Author/Year	Device Type	Material Used	Methodology	Key Results
[8], [9]	Lin & Yan (2012), Sirringhaus (2014)	FET / TFT	PANI-based organic material	Thin film deposition	Demonstrated transistor behavior
[4]	Gkoupidenis et al. (2015)	OECT	Conducting polymer (PANI/PEDOT)	Electrochemical modulation	High sensitivity, good modulation
[6]	Wang et al. (2019)	Thin film device	Crystalline PANI	Advanced material engineering	High conductivity (~160 S/cm)
[5]	Xia et al. (2012)	Thin films	Conducting polymer (PANI)	Solution processing	Low-cost fabrication
[3]	Torsi et al. (2013)	Organic TFT	Organic semiconductor	Device optimization	Good sensing capability

[1], [2]	MacDiarmid et al.	Conducting polymer	PANI	Chemical synthesis	High conductivity material

2.3 Identification of research gaps

Based on the reviewed literature, several important research gaps have been identified. While significant progress has been made in the development of PANI-based electronic devices, most studies are limited to a single device structure, such as transistors or thin films. There is a lack of comprehensive studies that investigate multiple device types using the same material system, which is essential for understanding the versatility of PANI in practical electronic applications.

Another major gap is the reliance on advanced fabrication techniques to achieve high-performance devices. Many reported results depend on sophisticated processes and controlled environments, which are not easily accessible in standard laboratory settings. This creates a need for research focused on simple, cost-effective fabrication methods that can be implemented using basic equipment.

Uniformity and quality of thin films also remain significant challenges. As highlighted in several studies, device performance is highly dependent on film morphology, thickness, and defect density [5], [6]. However, achieving consistent film quality using solution-based techniques is difficult, particularly under varying environmental conditions.

In addition, stability and reliability of organic devices are critical concerns. PANI-based devices are sensitive to environmental factors such as humidity and temperature, which can degrade their performance over time [1], [2]. Most studies do not adequately address long-term stability, limiting their practical applicability.

Furthermore, there is limited work on comparative analysis of different device structures fabricated using the same material. Such analysis is essential to identify the strengths and limitations of each device type and to determine their suitability for specific applications.

Finally, there is a lack of experimental studies demonstrating the fabrication and analysis of organic electronic devices at the undergraduate or small-scale research level. Addressing this gap is important for promoting accessibility and encouraging further research in organic electronics.

The present project aims to address these gaps by fabricating and analyzing multiple PANI-based devices, including MSM, MOS capacitor, and Thin Film Transistor structures, using simple and cost-effective fabrication techniques. This approach provides a broader understanding of the material behavior and highlights its potential for practical applications in flexible and low-cost electronics.

Chapter 3: System Design & Methodology

3.1 Design Considerations

The design of organic electronic devices in this project was carried out by carefully considering multiple engineering, environmental, and practical factors to ensure efficient performance, feasibility of fabrication, and long-term applicability. Since the project involves the use of Polyaniline (PANI) as an organic semiconductor and simple laboratory-based fabrication techniques, both technical and non-technical aspects were taken into account during the design phase.

Economic Considerations:
One of the primary motivations behind the design was to develop low-cost electronic devices. Conventional semiconductor fabrication involves expensive materials and cleanroom facilities, which are not always accessible. In contrast, Polyaniline (PANI) is relatively inexpensive and can be synthesized or procured easily. Additionally, fabrication techniques such as spin coating and basic photolithography significantly reduce production costs. The use of aluminum as the electrode material further contributes to cost reduction due to its availability and good electrical conductivity.

Environmental Considerations:
The environmental impact of the fabrication process was minimized by selecting materials and techniques that require lower energy consumption. Organic materials like PANI can be processed using solution-based methods at relatively low temperatures, unlike traditional semiconductor processes that involve high-temperature operations. This reduces energy usage and associated environmental impact. Furthermore, the use of minimal material quantities and controlled deposition helps in reducing waste generation.

Social Considerations:
The development of low-cost and flexible electronic devices has significant societal benefits. Such devices can be used in applications like wearable electronics, portable healthcare monitoring systems, and flexible sensors. By using simple fabrication methods, this project also promotes accessibility of advanced electronic device fabrication at the undergraduate level, enabling more students and researchers to explore this field.

Health and Safety Considerations:

During the fabrication process, various chemicals such as PANI solution, photoresist, and solvents are used. Proper safety measures were followed to ensure safe handling of these materials. Personal protective equipment such as gloves and masks were used, and all processes were carried out in well-ventilated laboratory conditions. Care was taken to avoid direct exposure to chemicals and to follow standard laboratory safety protocols.

Sustainability Considerations:

The project emphasizes sustainable fabrication practices by utilizing solution-processable materials and low-energy techniques. Organic electronics offer the potential for reduced environmental impact compared to traditional semiconductor technologies. Additionally, the use of flexible and lightweight materials supports the development of energy-efficient and portable electronic systems.

Technical Considerations:

Several technical factors were considered to ensure proper device functionality. These include achieving uniform thin films of PANI through controlled spin coating parameters, ensuring good adhesion between layers, and maintaining proper electrode spacing for effective current flow. The selection of Al_2O_3 as the dielectric material in the MOS capacitor and TFT was based on its good insulating properties and compatibility with the PANI layer. Careful attention was also given to minimizing defects such as pinholes and thickness variations, as these can significantly affect device performance.

Overall, the design integrates economic feasibility, environmental responsibility, safety, and technical efficiency, resulting in a practical and effective approach for the fabrication of PANI-based organic electronic devices.

3.2 Methodology

3.2.1 MSM

The Metal–Semiconductor–Metal (MSM) device was fabricated using Polyaniline (PANI) as the semiconductor material and Aluminum (Al) as the metal contacts. The fabrication process was carried out using standard laboratory techniques such as spin coating, photolithography, and metal deposition. The objective of this fabrication was to create a simple device structure to study the current–voltage (I–V) characteristics of PAN.

Materials Used

The following materials were used for the fabrication of the MSM device:

- Substrate (glass or silicon dummy substrate)
- Positive photoresist (AZ-1500)
- Aluminum (Al) for electrodes
- Polyaniline (PANI) solution
- Developer solution
- Solvents for cleaning (acetone, isopropyl alcohol, DI water)

Fabrication Steps

Step 1: Substrate Cleaning

The substrate was thoroughly cleaned to remove dust, grease, and organic contaminants. Cleaning was performed using acetone, isopropyl alcohol, and deionized water, followed by drying. This step is crucial to ensure proper adhesion of layers and to avoid defects in the device.

Step 2: Spin Coating of Photoresist

A uniform layer of positive photoresist (AZ-1500) was deposited on the substrate using a spin coater. The spin speed and time were controlled to achieve a uniform thin film. This layer is used to define the electrode pattern during lithography.

Step 3: Soft Baking

The coated substrate was soft baked at a moderate temperature to remove excess solvent from the photoresist and improve its adhesion to the substrate.

Step 4: UV Lithography

The substrate was exposed to ultraviolet (UV) light through a mask containing the interdigitated

electrode pattern. The exposed regions of the photoresist underwent chemical changes, making them soluble in the developer solution.

Step 5: Development

The substrate was immersed in a developer solution to remove the exposed photoresist, thereby transferring the electrode pattern onto the substrate.

Step 6: Metal Deposition

Aluminum (Al) was deposited onto the patterned substrate using a deposition technique such as thermal evaporation or sputtering. This forms the metal electrodes of the MSM structure.

Step 7: Lift-off Process

The remaining photoresist along with unwanted metal was removed using a suitable solvent. This process leaves behind only the desired metal electrode pattern on the substrate.

Step 8: Deposition of PANI Layer

A thin film of Polyaniline (PANI) was spin-coated over the electrode structure. This forms the semiconductor layer connecting the metal electrodes. The film was then dried to ensure proper adhesion and uniformity.

Final Device Structure

The fabricated MSM device consists of two aluminum electrodes separated by a thin layer of PANI, forming a Metal–Semiconductor–Metal configuration. The electrodes are typically arranged in an interdigitated pattern to increase the effective contact area and improve current flow.

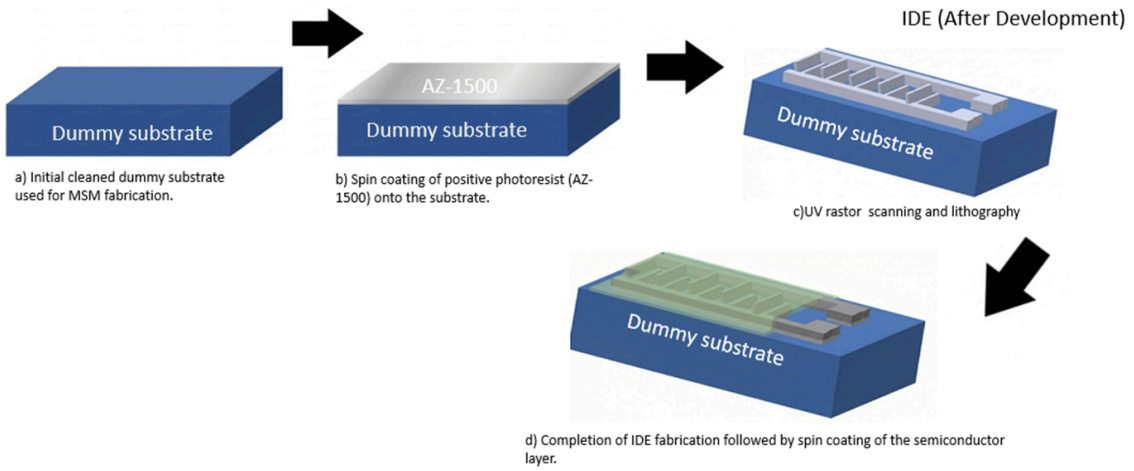
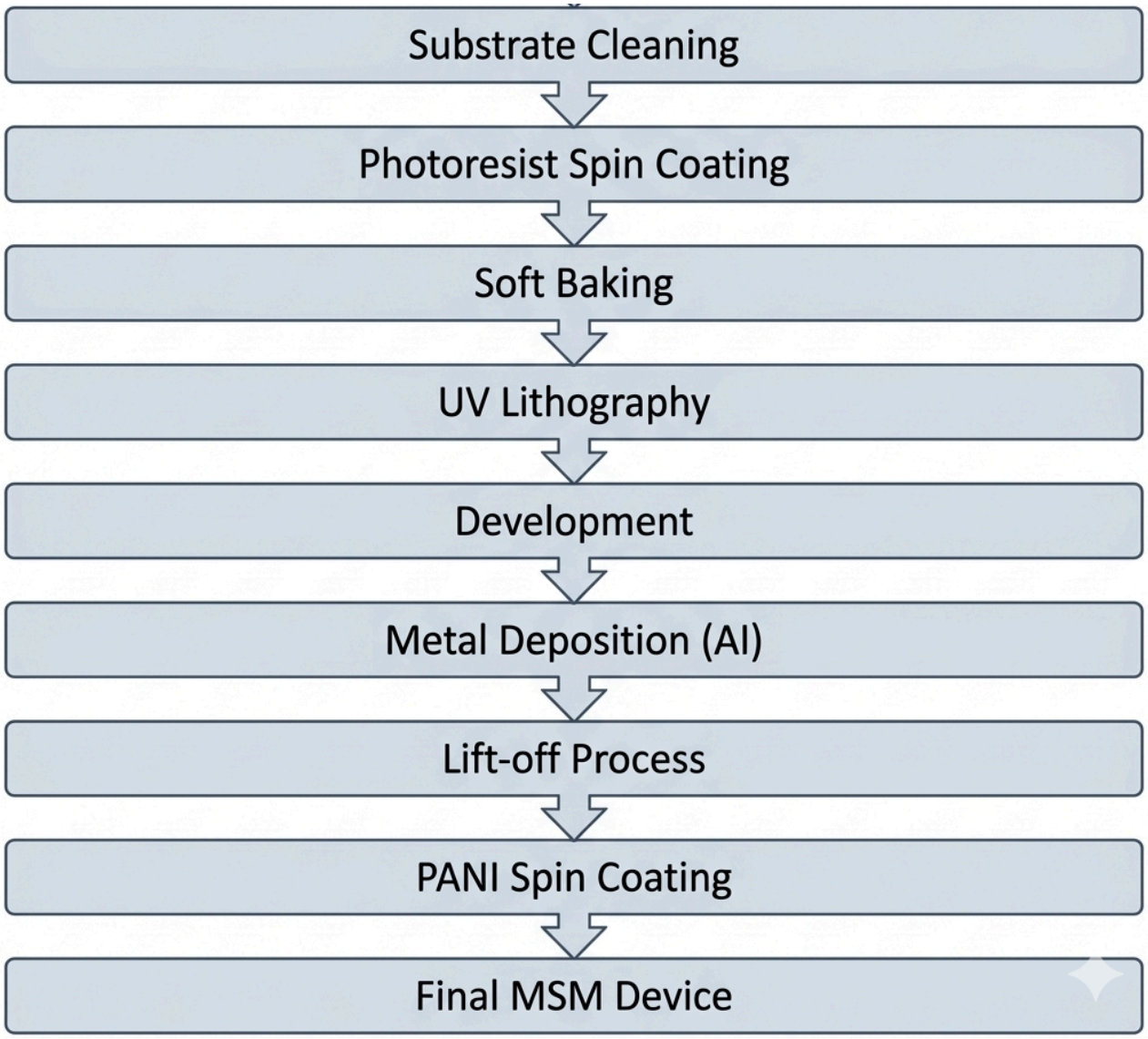


Fig 3.1: Fabrication Flow of MSM

Key Fabrication Considerations

- Uniformity of the PANI thin film is critical for consistent electrical behavior
- Proper alignment during lithography ensures accurate electrode pattern formation
- Clean substrate surface improves adhesion and reduces defects
- Controlled deposition of metal ensures good electrical contact

Flowchart of MSM Fabrication Process



3.2.2 MOS CAPACITOR

The MOS capacitor was fabricated using Polyaniline (PANI) as the semiconductor layer, Aluminum Oxide (Al_2O_3) as the dielectric layer, and Aluminum (Al) as the metal contacts. The fabrication process involved multiple thin film deposition techniques, including DC sputtering for metal deposition, RF sputtering for dielectric formation, and spin coating for the PANI layer.

- Materials Used

• Substrate (glass or silicon dummy substrate)
• Aluminum (Al) for top and bottom electrodes
• Polyaniline (PANI) solution
• Aluminum Oxide (Al_2O_3) target
• Solvents (acetone, isopropyl alcohol, DI water)

- Fabrication Steps

Step 1: Substrate Cleaning
The substrate was cleaned using acetone, isopropyl alcohol, and deionized water to remove contaminants. Proper cleaning ensures good adhesion of deposited layers and minimizes defects.

Step 2: Bottom Metal Deposition (DC Sputtering / Thermal Evaporation)
A thin layer of Aluminum (Al) was deposited on the cleaned substrate to form the bottom electrode. This was achieved using DC sputtering, where aluminum atoms are ejected from a target and deposited onto the substrate, ensuring uniform coverage and good adhesion.
In some cases, thermal evaporation can also be used as an alternative method, where aluminum is heated and deposited in vacuum conditions. This step forms one plate of the MOS capacitor.

Step 3: Deposition of PANI Semiconductor Layer (Spin Coating)
Polyaniline (PANI) solution was deposited onto the bottom electrode using the spin coating technique. A controlled amount of solution was dropped onto the substrate and spun at 1500 rpm for 60 seconds to achieve a uniform thin film.

After spin coating, the substrate was dried for 10 minutes to remove residual solvent and ensure proper film formation. The uniformity of this layer is critical, as it directly affects device performance.

Step 4: Dielectric Layer Deposition (RF Sputtering)

A thin insulating layer of Aluminum Oxide (Al_2O_3) was deposited over the PANI layer using RF sputtering. This technique is suitable for insulating materials and ensures the formation of a uniform, dense dielectric layer.

During RF sputtering, plasma is generated using a high-frequency alternating field, which causes atoms from the Al_2O_3 target to be deposited onto the substrate. The dielectric layer plays a crucial role in charge storage and electrical isolation between the metal layers.

Step 5: Top Metal Deposition (DC Sputtering / Thermal Evaporation)

Finally, a top Aluminum (Al) electrode was deposited over the dielectric layer using DC sputtering. This completes the MOS capacitor structure. Similar to the bottom electrode, thermal evaporation can also be used as an alternative deposition technique depending on equipment availability.

Final Device Structure

The fabricated MOS capacitor has the following layered structure:

Al (Top Metal) / Al_2O_3 (Dielectric) / PANI (Semiconductor) / Al (Bottom Metal)

This structure enables the study of capacitance variation with applied voltage and analysis of charge behavior at the dielectric–semiconductor interface.

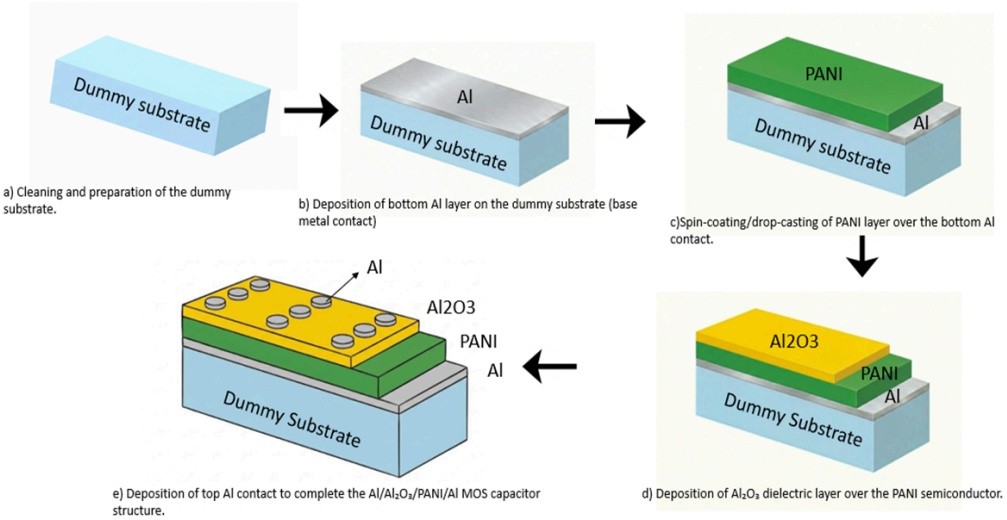


Fig 3.2 Fabrication flow MOS Capacitor

Key Fabrication Considerations

- Uniform spin-coated PANI layer is essential for consistent device behavior
- RF sputtering ensures high-quality dielectric deposition with good insulation properties
- Proper metal deposition using DC sputtering improves electrode conductivity and adhesion
- Thickness of dielectric layer directly affects capacitance
- Clean interfaces between layers reduce defects and improve performance

Flowchart of MOS Capacitor Fabrication



3.2.3 Thin Film Transistor (TFT)

The Thin Film Transistor (TFT) was fabricated using Polyaniline (PANI) as the semiconductor layer, Aluminum Oxide (Al_2O_3) as the dielectric layer, and Aluminum (Al) as the metal contacts. The fabrication process followed a sequential approach involving oxide formation, photolithography, thin film deposition, and spin coating techniques.

Materials Used

- Substrate (silicon/glass)
- Aluminum (Al)
- Aluminum Oxide (Al_2O_3)
- Polyaniline (PANI) solution
- Positive photoresist (AZ-1500)
- Developer solution
- Cleaning solvents (acetone, IPA, DI water)

Fabrication Steps

Step 1: Substrate Cleaning
The substrate was cleaned using acetone, isopropyl alcohol, and deionized water to remove contaminants and ensure proper adhesion of layers.

Step 2: Oxide Layer Formation (Al_2O_3 Deposition)
An initial oxide layer was formed on the substrate using RF sputtering to deposit Aluminum Oxide (Al_2O_3). This layer acts as the dielectric base for the TFT structure and provides electrical insulation.

Step 3: Photolithography for Electrode Patterning (IDE Formation)
A layer of positive photoresist (AZ-1500) was spin-coated onto the substrate and soft baked. UV lithography was performed using a mask to define the interdigitated electrode (IDE) pattern for source and drain. After exposure, the substrate was developed to transfer the pattern.

Step 4: Aluminum Deposition for Electrodes (DC Sputtering)
Aluminum (Al) was deposited using DC sputtering over the patterned substrate to form the interdigitated source and drain electrodes.

Step 5: Lift-off Process

The unwanted aluminum along with the remaining photoresist was removed using a solvent, leaving behind the desired IDE electrode pattern.

Step 6: Deposition of PANI Semiconductor Layer (Spin Coating)

Polyaniline (PANI) solution was deposited over the electrode structure using spin coating. The substrate was spun at 1500 rpm for 60 seconds to achieve a uniform thin film.

After coating, the film was dried for 10 minutes to remove residual solvent and ensure proper film formation. This layer acts as the active channel of the TFT.

Step 7: Top Dielectric Deposition (Al_2O_3 using RF Sputtering)

A second layer of Aluminum Oxide (Al_2O_3) was deposited over the PANI layer using RF sputtering. This layer serves as the gate dielectric in the TFT structure.

Step 8: Final Metal Deposition (Gate Electrode – Al)

Finally, Aluminum (Al) was deposited using DC sputtering (or thermal evaporation) to form the gate electrode on top of the dielectric layer.

Final Device Structure

The fabricated TFT consists of a layered structure:

Al (Gate) / Al_2O_3 (Dielectric) / PANI (Semiconductor) / Al (Source & Drain IDE) / Al_2O_3 (Base Oxide)

This structure enables modulation of current between source and drain terminals through the applied gate voltage.

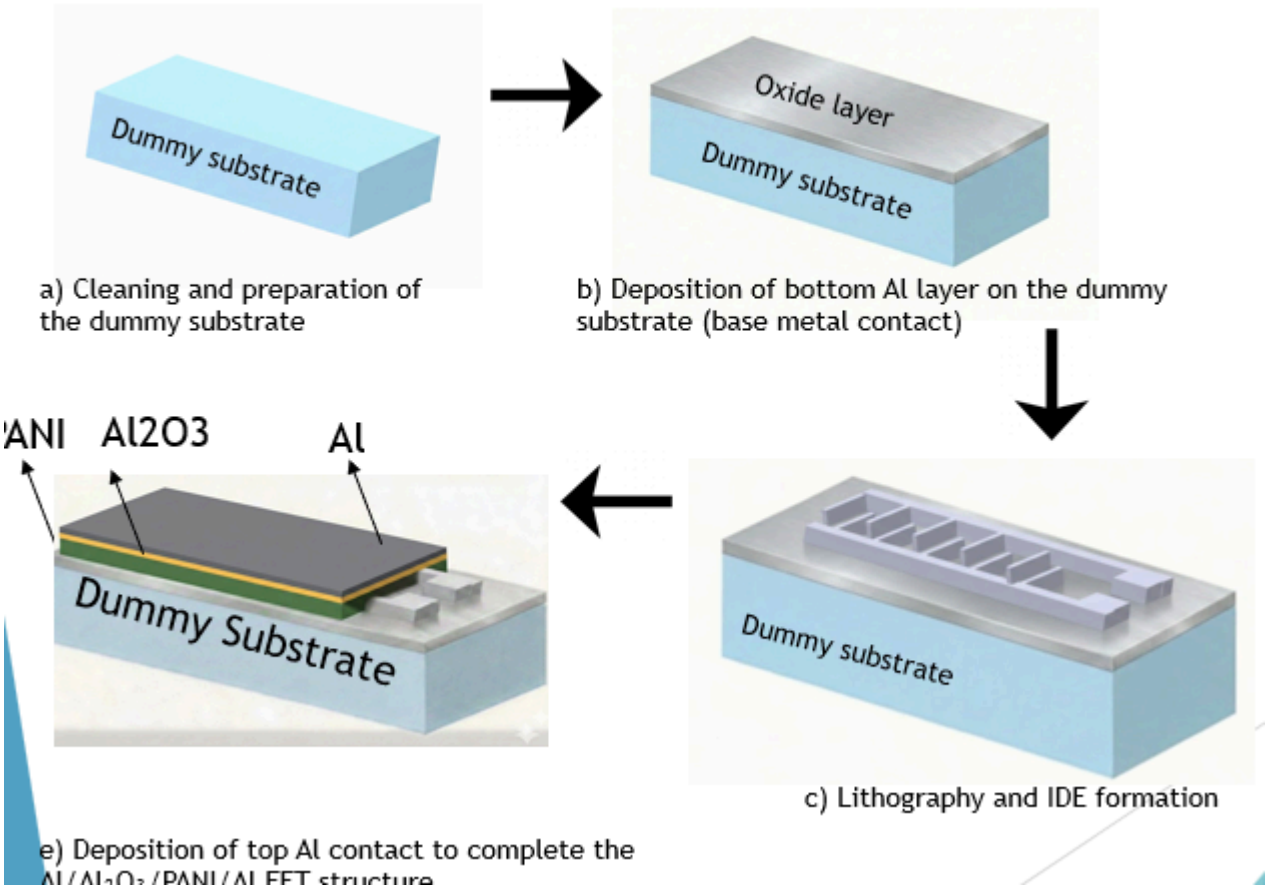
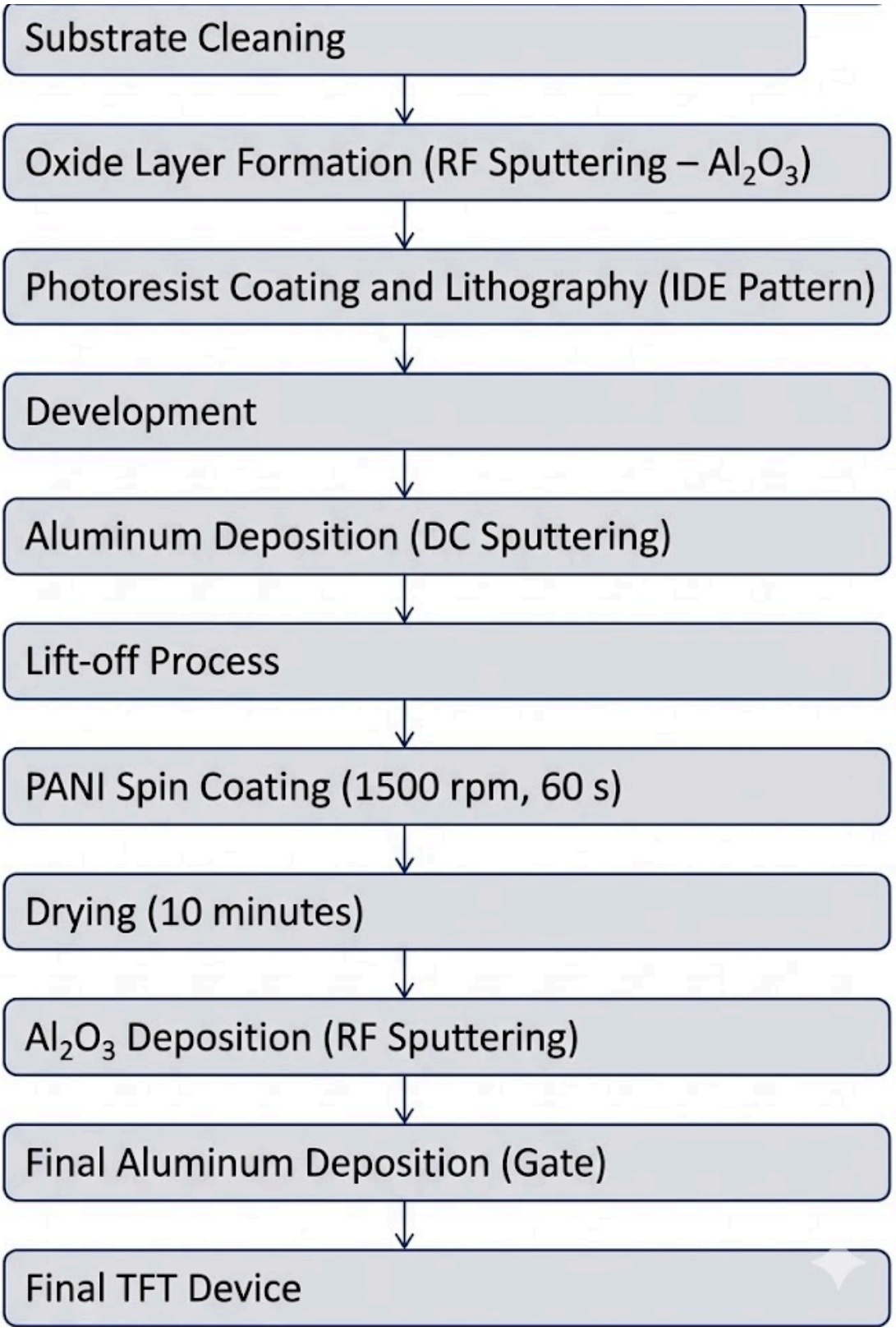


Fig 3.3 Fabrication flow TFT

Key Fabrication Considerations

- Proper IDE patterning is essential for channel formation
- Uniform PANI coating ensures consistent electrical behavior
- Dielectric quality directly affects gate control
- Clean interfaces between layers reduce defects
- Alignment and deposition accuracy impact device performance

Flowchart of TFT Fabrication Process



3.3 Modelling

3.3.1 Modelling of MSM Device

The Metal–Semiconductor–Metal (MSM) structure is one of the simplest and most important device configurations used to study charge transport in semiconductor materials. It consists of a semiconductor layer (Polyaniline in this project) sandwiched between two metal contacts, forming two back-to-back Schottky junctions.

Unlike MOS-based devices, which rely on charge modulation at the oxide–semiconductor interface, the operation of an MSM device is governed by the behavior of the Schottky barriers formed at the metal–semiconductor interfaces. When a voltage is applied across the two metal electrodes, one junction becomes forward-biased while the other becomes reverse-biased. The overall current flow through the device is therefore controlled by the reverse-biased junction, which acts as the limiting factor.

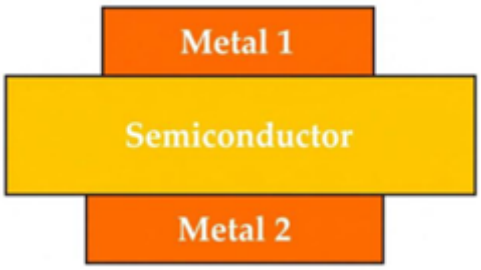


Fig. 3.4: Structure of a MSM

Working Principle

At low applied voltages, the current through the MSM device is very small due to the presence of the reverse-biased Schottky barrier. This barrier prevents charge carriers from easily crossing the metal–semiconductor interface. As the applied voltage increases, the electric field across the semiconductor also increases, enabling charge carriers to overcome or tunnel through the barrier.

The dominant current transport mechanisms include:

- Thermionic emission
- Thermionic field emission
- Field emission (tunneling)

As a result, the current increases gradually at first and then more rapidly at higher voltages.

I–V Characteristics

The current–voltage (I–V) characteristics of an MSM device are typically non-linear due to the presence of Schottky barriers. At very low voltages, the device may exhibit quasi-linear (ohmic-like) behavior if the contacts are nearly ohmic. However, in most practical cases, the current follows an exponential trend.

The current through a Schottky junction can be expressed as:

$$I = I_s \left(e^{\frac{qV}{nkT}} - 1 \right)$$

where:
Is = saturation current
q = electronic charge
V = applied voltage
n = ideality factor
k = Boltzmann constant
T = temperature

Since the MSM device consists of two back-to-back Schottky diodes, the overall I–V characteristic appears symmetric with respect to voltage polarity.

Leakage Current in MSM Devices

In practical MSM devices, a small amount of current flows even under reverse-bias conditions. This is known as leakage current. Ideally, the reverse-biased junction should block current completely; however, due to real-world imperfections, a finite current is always present.

Leakage current arises due to:

- Thermionic emission over the barrier
- Tunneling through the barrier
- Defect-assisted carrier transport

- Surface states and traps

In the I–V curve, leakage current is observed as a small but non-zero current in the low-voltage region. As the voltage increases, the effective barrier height decreases due to electric field effects (Schottky barrier lowering), leading to a gradual increase in leakage current before the onset of strong conduction.

Role and Importance of MSM Structures

MSM devices play a significant role in both electronic and optoelectronic applications due to their simple structure and fast response. Their symmetric configuration and low capacitance enable high-speed operation, making them suitable for photodetectors, optical sensors, and switching devices.

Additionally, MSM structures are widely used as test devices to analyze semiconductor properties such as carrier transport mechanisms, barrier height, and interface quality. In this project, the MSM device serves as a fundamental structure to study the electrical behavior of Polyaniline and evaluate its suitability for electronic applications.

Overall, the modelling of the MSM device provides a clear understanding of charge transport through metal–semiconductor interfaces and supports the analysis of experimental I–V characteristics obtained from the fabricated device.

3.3.2 Modelling of MOS CAPACITOR

The Metal–Oxide–Semiconductor (MOS) capacitor is a fundamental device structure used to study charge distribution and electrical behavior at the semiconductor–dielectric interface. In this project, an organic MOS capacitor was fabricated using Polyaniline (PANI) as the semiconductor, Aluminum Oxide (Al_2O_3) as the dielectric, and Aluminum (Al) as the metal electrode.

Unlike a conventional capacitor, whose capacitance remains constant, the capacitance of a MOS capacitor varies with the applied voltage due to changes in charge distribution within the semiconductor. This voltage-dependent behavior makes it an essential structure for analyzing semiconductor properties and forms the basis for transistor operation.

Structure of MOS Capacitor

The MOS capacitor consists of three layers:

- Metal (Al) – top electrode
- Dielectric (Al_2O_3) – insulating layer
- Semiconductor (PANI) – active layer

This forms a Metal–Oxide–Semiconductor (MOS) structure, where the applied voltage at the metal controls the charge distribution in the semiconductor.

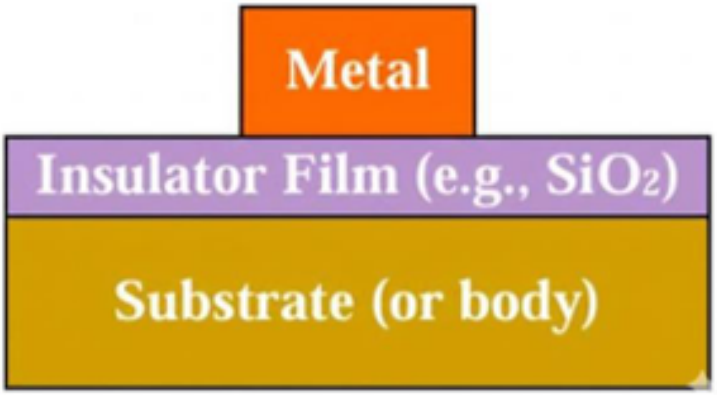


Fig 3.5 Structure of MOS Capacitor

Capacitance Equation

The capacitance of the MOS capacitor is given by:

$$C = \frac{\varepsilon_{ox} A}{t_{ox}}$$

where:
εox = permittivity of oxide
A = area of capacitor
tox = thickness of oxide layer

Operating Regions of MOS Capacitor

The operation of the MOS capacitor can be divided into three regions depending on the applied gate voltage.

Accumulation Region

When a negative voltage is applied to the gate (for p-type behavior), majority carriers (holes in PANI) are attracted towards the oxide–semiconductor interface. This results in a high concentration of charge carriers near the interface.

In this region:

- Capacitance is maximum
- Capacitance ≈ oxide capacitance ($C \approx Cox$)
- No depletion layer is formed

The device behaves similar to a conventional capacitor.

Depletion Region

When a small positive voltage is applied, majority carriers are repelled away from the interface, leaving behind a region depleted of mobile charge carriers.

This leads to:

- Formation of depletion layer
- Increase in effective dielectric thickness
- Reduction in capacitance

The total capacitance is given by:

$$C = \frac{C_{ox} \cdot C_{dep}}{C_{ox} + C_{dep}}$$

where *Cdep* is the depletion capacitance.

Inversion Region

At higher positive voltages, minority carriers begin to accumulate near the interface. In conventional silicon devices, this leads to strong inversion and formation of a conductive channel.

However, in organic semiconductors like PANI:

- Inversion is weaker
- Charge transport is dominated by traps and hopping conduction
- Transition between regions is gradual

At low frequency:

- Capacitance increases again toward *Cox*

At high frequency:

- Capacitance saturates at a lower value due to slow response of minority carriers

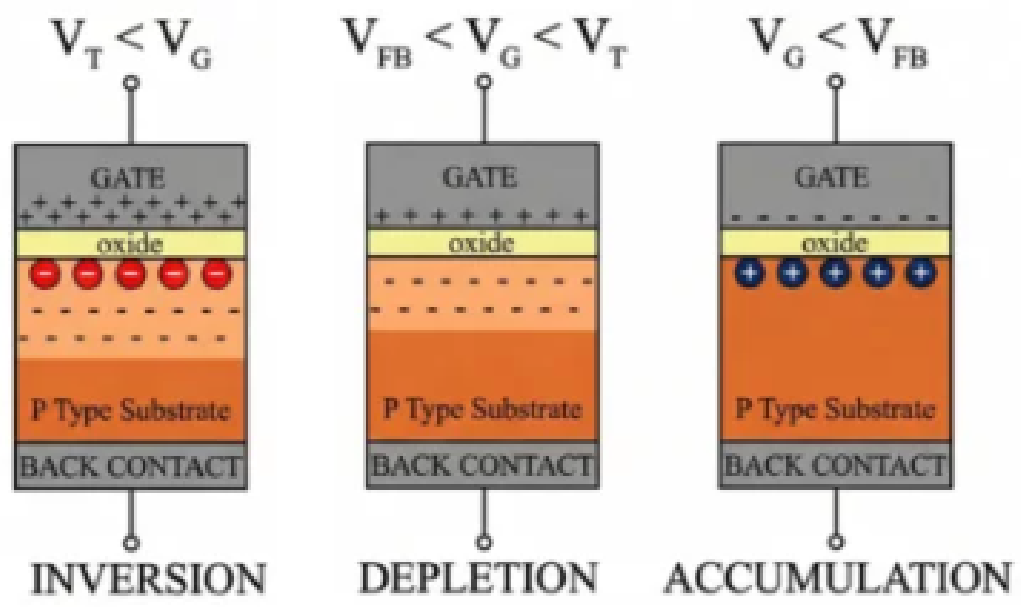


Fig 3.6 Regions of operations of MOS Cap

Threshold Voltage

The threshold voltage (*VT*) is the minimum gate voltage required to initiate inversion.

It is given by:

$$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{2\varepsilon_s q N_A (2\phi_F)}}{C_{ox}}$$

This parameter depends on:

- Doping concentration
- Oxide thickness
- Interface charges

Flat-Band Voltage

Flat-band voltage (*VFB*) is the voltage at which energy bands in the semiconductor are flat, indicating no charge accumulation or depletion.

$$V_{FB} = \phi_{MS} - \frac{Q_{ox}}{C_{ox}}$$

It is important for analyzing:

- Interface quality
- Oxide charges
- Device reliability

C–V Characteristics

The capacitance–voltage (C–V) characteristics of a MOS capacitor provide detailed information about device behavior.

Key observations:

• High capacitance in accumulation
• Decreasing capacitance in depletion
• Frequency-dependent behavior in inversion

Types of C–V curves:

- Low-frequency C–V: Minority carriers respond → capacitance returns to *Cox*
- High-frequency C–V: Minority carriers cannot respond → capacitance saturates

These curves help in extracting:

- Oxide thickness
- Doping concentration
- Interface trap density

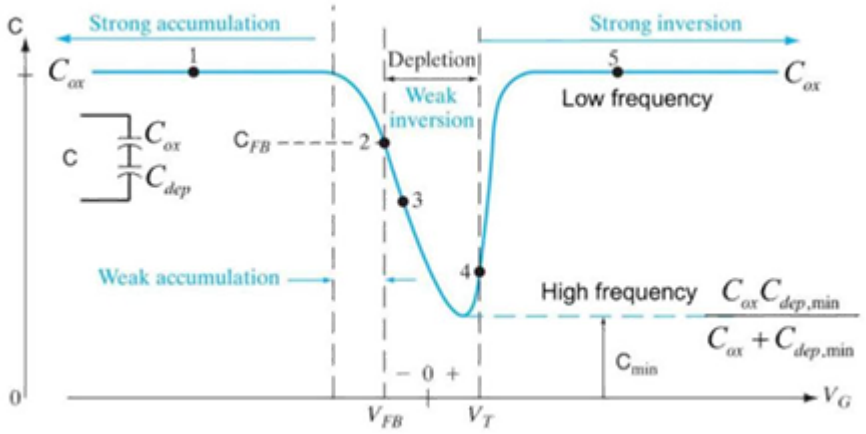


Fig. 3.7: C-V Characteristics of a p-si based MOS Capacitor

Organic MOS Capacitor Behavior

In organic MOS capacitors (using PANI):

• Charge transport occurs via hopping mechanisms
• Higher trap density leads to hysteresis
• Transitions between regions are smoother
• Strong inversion is often not clearly observed

These characteristics differentiate organic MOS capacitors from conventional silicon-based devices.

Role of MOS Capacitor in TFT

The MOS capacitor serves as the fundamental building block of Thin Film Transistors (TFTs). The same principle of charge modulation at the semiconductor–dielectric interface is used in TFT operation.

Parameters such as:

- Oxide capacitance (*Cox*)
- Threshold voltage
- Interface quality

are first extracted using MOS capacitor analysis and then used to design efficient TFT devices.

Importance of MOS Capacitor

The MOS capacitor is essential because:

- It helps in understanding semiconductor–dielectric interface behavior
- It allows extraction of key electrical parameters
- It acts as a test structure before transistor fabrication
- It is fundamental to the operation of MOSFETs and TFTs

Overall, the modelling of the MOS capacitor provides critical insights into charge modulation, capacitance variation, and interface properties, which are essential for analyzing and optimizing organic electronic devices.

3.3.3 Modelling of Thin Film Transistor

The Thin Film Transistor (TFT) is a three-terminal device in which the current flowing between the source and drain electrodes is controlled by the voltage applied at the gate terminal. In this project, an organic TFT was fabricated using Polyaniline (PANI) as the semiconductor layer and Aluminum Oxide (Al_2O_3) as the dielectric.

The operation of the TFT is fundamentally based on the same principle as the MOS capacitor—modulation of charge at the semiconductor–dielectric interface. However, unlike the MOS capacitor, the TFT allows controlled current flow, making it an active switching device.

Structure of TFT

The fabricated TFT consists of the following layers:

- Gate electrode (Al)
- Dielectric layer (Al_2O_3)
- Semiconductor layer (PANI)
- Source and Drain electrodes (Al – interdigitated)

When a voltage is applied at the gate, an electric field is created across the dielectric, which controls the charge carrier concentration in the semiconductor channel between source and drain.

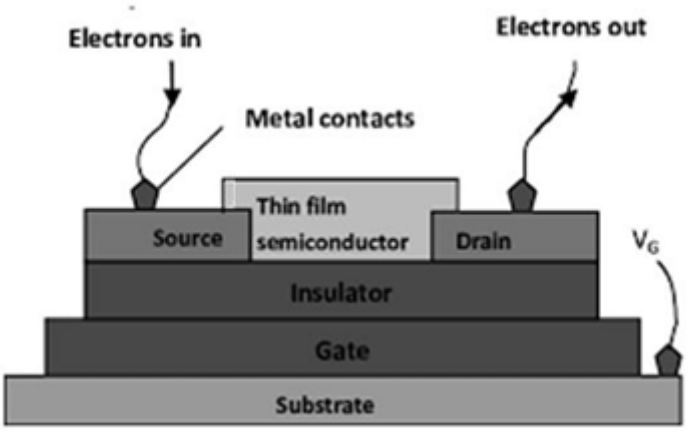


Fig 3.8 Structure of TFT

Working Principle

When a gate voltage (*VGS*) is applied, it modulates the charge carrier density at the semiconductor–dielectric interface:

• At low gate voltage → no channel forms → no current
• At moderate voltage → partial channel formation → small current
• At high voltage → strong channel → high current

Thus, the TFT acts as a voltage-controlled current device.

Operating Regions of TFT

Cutoff Region

When:

$$V_{GS} < V_T$$

No conducting channel is formed, and the drain current is approximately zero.

Linear (Ohmic) Region

When:

$$V_{GS} > V_T \quad \text{and} \quad V_{DS} < (V_{GS} - V_T)$$

The device behaves like a resistor, and current increases linearly with drain voltage.

$$I_D = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Saturation Region

When:

$$V_{DS} \geq (V_{GS} - V_T)$$

The channel gets pinched off near the drain, and the current becomes constant.

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

Transfer and Output Characteristics

Transfer Characteristics (ID vs VGS):

- Shows how current varies with gate voltage
- Used to determine threshold voltage and mobility

Output Characteristics (ID vs VDS):

- Shows different operating regions
- Linear → Saturation transition

These graphs are essential for analyzing device performance.

Capacitance and Channel Formation

The oxide capacitance (*Cox*) plays a crucial role in determining the channel charge:

$$Q = C_{ox}(V_{GS} - V_T)$$

Higher capacitance leads to stronger gate control and better device performance.

HOMO–LUMO Concept in Organic TFT

In organic semiconductors like Polyaniline (PANI), charge transport is governed by molecular energy levels rather than continuous energy bands.

• **HOMO (Highest Occupied Molecular Orbital):**
Equivalent to the valence band → hole transport

• **LUMO (Lowest Unoccupied Molecular Orbital):**
Equivalent to the conduction band → electron transport

The energy difference between HOMO and LUMO is the bandgap, which determines electrical and optical properties.

For PANI (p-type material):

- Charge transport mainly occurs via holes in the HOMO level
- Gate voltage controls hole accumulation at the interface

Proper alignment of HOMO level with metal work function:

- Reduces injection barrier
- Improves current flow
- Enhances device performance

Organic TFT Behaviour

Compared to conventional silicon TFTs:

• Charge transport occurs via hopping mechanism
• Mobility is lower

• Presence of traps causes hysteresis
• Threshold voltage shifts may occur
• Switching is slower but sufficient for low-cost electronics

Importance of TFT Modelling

TFT modelling helps in:

• Understanding device operation
• Predicting electrical characteristics
• Extracting parameters like mobility and threshold voltage
• Optimizing fabrication and material selection

Overall, the modelling of the Thin Film Transistor provides a complete understanding of its electrical behavior and highlights the role of organic materials like PANI in flexible and low-cost electronic applications.

3.4 Tool and Technique Selection

The selection of tools and fabrication techniques in this project was based on achieving uniform thin films, reliable device performance, and compatibility with organic semiconductor processing. Each technique was chosen to optimize specific aspects such as film quality, interface properties, reproducibility, and electrical characterization.

Wafer Preparation and Cutting

The initial step involved cutting the wafer into smaller square pieces using a diamond cutter. This was necessary to obtain manageable substrate sizes suitable for laboratory-scale fabrication. Smaller substrates also ensure uniform deposition, ease of handling, and efficient utilization of materials.



Fig3.9 : wafer cut into pieces

Substrate Cleaning

Proper cleaning of substrates is critical for achieving high-quality thin films and reliable device performance. A standard laboratory cleaning method was used instead of RCA cleaning to maintain simplicity while ensuring sufficient cleanliness.

The substrates were:

- Rinsed with distilled water
- Cleaned with detergent to remove grease and particles
- Washed with acetone and isopropyl alcohol (IPA) to remove organic contaminants
- Dried using a hot air blower

This method was chosen because:

- It effectively removes contaminants
- It is simple and cost-effective
- It provides adequate surface quality for polymer-based devices

Clean substrates ensure good adhesion of PANI films and minimize defects such as pinholes and non-uniform coatings.

Sputtering Technique

Sputtering was selected as the primary thin-film deposition technique due to its ability to produce uniform, high-quality films with good adhesion. The process involves plasma generation using an inert gas (argon), where energetic ions bombard a target material, causing atoms to be ejected and deposited onto the substrate.

Sputtering was preferred because:
• It provides excellent film uniformity
• It allows controlled thickness deposition
• It is suitable for both metals and dielectrics
• It ensures strong adhesion between layers

RF Sputtering for Al_2O_3 Deposition

RF sputtering was specifically used for depositing Aluminum Oxide (Al_2O_3), which is an insulating material. Unlike DC sputtering, RF sputtering prevents charge accumulation on insulating targets by using an alternating electric field.

It was chosen because:
• It enables deposition of high-quality dielectric films
• It avoids arcing and charge buildup
• It ensures uniform and dense oxide layers
• It is ideal for MOS and TFT dielectric formation

The addition of a magnetron improves plasma confinement and enhances deposition efficiency, resulting in better film quality.

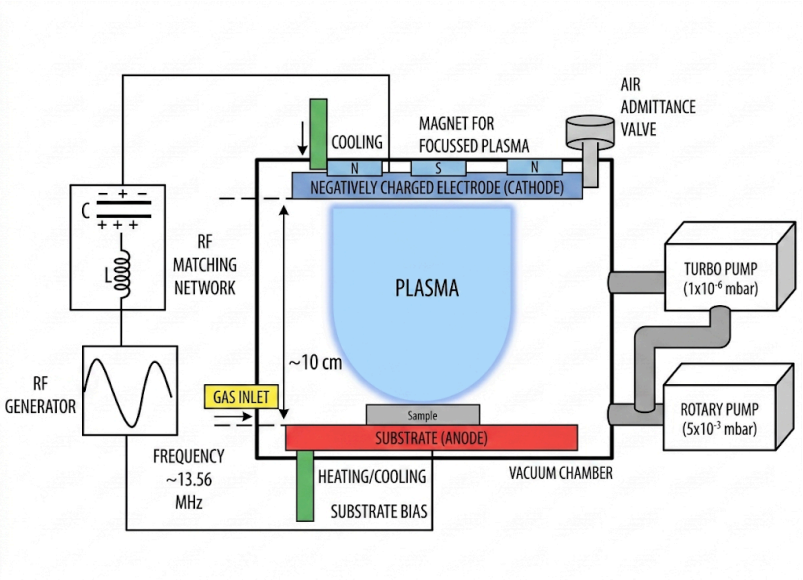


Fig. 3.10 : Set-up for deposition of Al2O3 layer over p-si substrate.

DC Sputtering for Metal Contacts (Al)

DC sputtering was used for depositing Aluminum (Al) electrodes, as it is highly suitable for conductive materials.

It was selected because:

• It provides uniform and high-purity metal films

• It offers good control over thickness

• It is cost-effective for metal deposition

• It ensures strong electrical contacts

In this process, argon ions bombard the metal target, and the ejected atoms deposit onto the substrate, forming the electrode layer.

Thermal Evaporation for Metal Deposition

Thermal evaporation was also used as an alternative method for depositing metal contacts. In this process, metal is heated in a high-vacuum chamber until it evaporates and condenses onto the substrate.

This technique was chosen because:

• It produces highly pure metal films

• It operates under high vacuum, reducing contamination

• It provides smooth and uniform coatings
• It is suitable for thin metal layers

Typical conditions include:
• Vacuum: 10^{-6} to 10^{-7} Torr
• Deposition rate: 1–5 Å/s

Thermal evaporation complements sputtering by offering flexibility in deposition methods.

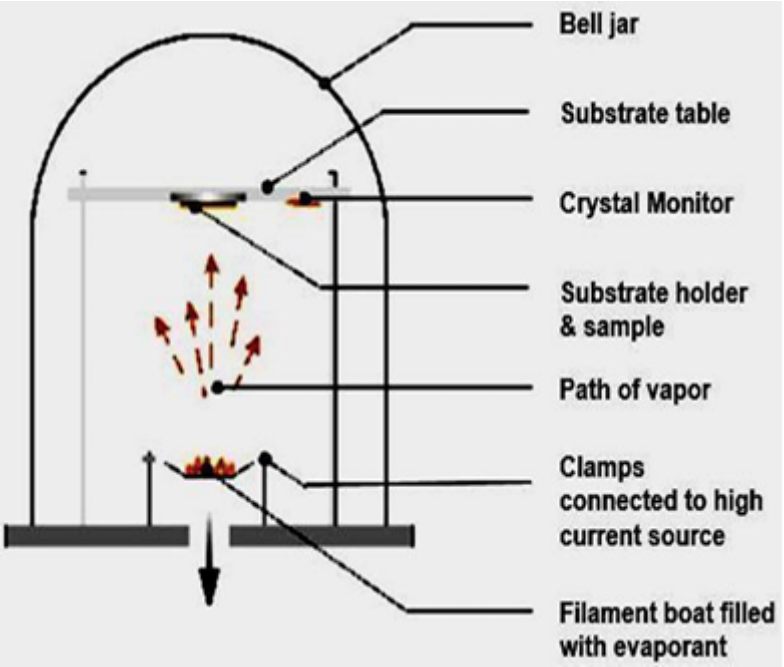


Fig 3.11 Setup for Thermal Evaporation for metal deposition

Spin Coating of PANI Thin Films

Spin coating was used for depositing Polyaniline (PANI) thin films due to its simplicity and effectiveness in forming uniform coatings.

Process details:
• Spin speed: 1500 rpm
• Time: 60 seconds
• Drying: 10 minutes after each layer
• Repeated ~20 times for desired thickness

This technique was chosen because:
• It provides uniform thin films
• It is low-cost and easy to implement

• It allows control over thickness via multiple coatings
• It minimizes defects like bubbles and cracks

The multi-layer deposition approach ensures better film uniformity and improved electrical performance.

Photolithography

Photolithography was used to define electrode patterns (such as interdigitated electrodes in MSM and TFT devices).

It was selected because:
• It provides high precision patterning
• It allows accurate control of device dimensions
• It is essential for microfabrication

This step ensures proper channel formation and device functionality.

Synthesis of Polyaniline (PANI)

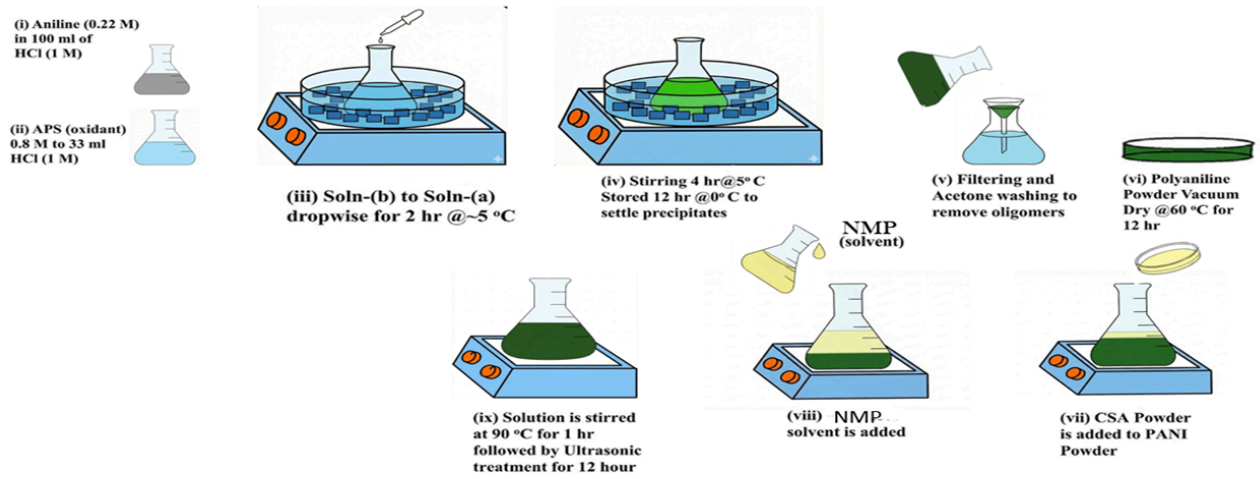
Polyaniline was synthesized using chemical oxidative polymerization due to its simplicity and effectiveness in producing high-quality conducting polymers.

Key steps:
• Polymerization at low temperature (0–5°C)
• Use of APS as oxidizing agent
• Controlled stirring and reaction time
• Washing and purification
• Drying and dispersion in NMP

This method was chosen because:
• It produces high molecular weight PANI
• It ensures better electrical properties
• It provides good film-forming capability
• It allows easy processing for thin-film deposition

The use of NMP improves solubility and ensures uniform coating during spin coating.

Fig 3.12 Synthesis of Polyaniline using NMP as a solvent



Software Tool: OriginPro

OriginPro software was used for plotting and analyzing I–V and C–V characteristics of the fabricated devices.

The process involved:

• Importing voltage and capacitance/current data

• Plotting graphs (I–V and C–V curves)

• Performing curve fitting and analysis

OriginPro was chosen because:

• It provides high-quality scientific plotting

• It supports advanced data analysis (fitting, derivatives, integration)

• It allows extraction of parameters like threshold voltage and capacitance

• It enables export of publication-quality graphs

This makes it highly suitable for semiconductor device characterization.

ORIGINPRO®
The Ultimate Software for Graphing & Analysis

Fig 3.13 Origin Pro

Overall Justification

The combination of sputtering, thermal evaporation, spin coating, and photolithography ensured precise fabrication of multilayer device structures. The synthesis of PANI enabled the use of an organic semiconductor, while OriginPro provided accurate analysis of electrical characteristics.

Together, these tools and techniques ensured:

- High-quality thin films
- Reliable device fabrication
- Accurate electrical characterization
- Reproducibility of results

Thus, the selected methodologies were well-suited for the successful fabrication and analysis of MSM, MOS capacitor, and TFT devices using Polyaniline.

Chapter 4: Implementation & Testing

4.1 Development Details

The implementation of the project was carried out in a systematic manner to fabricate three different organic electronic devices using Polyaniline (PANI) as the active semiconductor material. The complete development process included substrate preparation, thin film deposition, pattern formation, device assembly, and electrical testing. Each stage was carefully executed to ensure proper layer formation and reliable device operation.

Initially, the substrates were cleaned using acetone, isopropyl alcohol, and deionized water to remove dust, grease, and organic contaminants. Clean substrates were necessary to improve adhesion of deposited layers and reduce fabrication defects.

For the MSM device, photolithography was used to define the interdigitated electrode pattern. A positive photoresist layer was spin coated and exposed to UV light through a mask. After development, aluminum electrodes were deposited using sputtering or evaporation. Finally, PANI was spin coated over the patterned electrodes to form the semiconductor path between contacts.

For the MOS capacitor, the bottom aluminum electrode was first deposited on the substrate. A uniform PANI thin film was then deposited using spin coating. After drying, an Al_2O_3 dielectric layer was formed using RF sputtering. A top aluminum electrode was deposited to complete the capacitor structure.

For the Thin Film Transistor (TFT), a base oxide layer was first deposited. Source and drain electrodes were patterned using lithography followed by aluminum deposition. A PANI layer was spin coated to form the active channel region. An additional Al_2O_3 dielectric layer was deposited, followed by formation of the gate electrode.

After fabrication, all devices were electrically characterized using available laboratory instruments. Current-voltage and capacitance-voltage responses were recorded and analyzed using OriginPro software.



Fig. 4.1: Thermal Evaporation with the boat



Fig 4.2 weight measurement of Polyaniline



Fig 4.3 Substrate before DC Sputtering

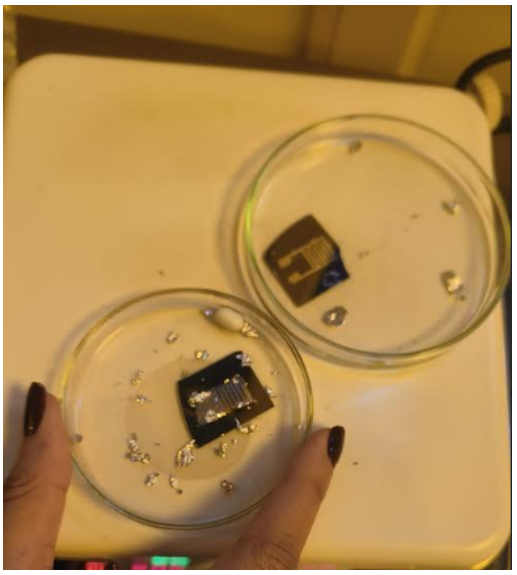


Fig 4.4 Substrate during Lift off Process

4.2 Experimental Setup

The experimental setup consisted of fabrication equipment, thin film deposition systems, and characterization instruments required for the successful realization of the devices.

Hardware Components

- Spin Coater for uniform deposition of PANI thin films
- UV Lithography setup for pattern transfer
- RF Sputtering unit for Al_2O_3 dielectric deposition
- DC Sputtering / Thermal Evaporation unit for aluminum deposition
- Hot Plate for soft baking and drying
- Probe Station for electrical testing
- Multimeter / Source Meter for I-V measurements
- LCR Meter for capacitance measurements

Software Components

- OriginPro for graph plotting and analysis
- System control software associated with deposition instruments

Materials Used

- Polyaniline (PANI) solution
- Aluminum target
- Aluminum Oxide target
- AZ-1500 photoresist
- Developer solution
- Acetone
- Isopropyl Alcohol
- Deionized Water

The fabricated devices were tested under normal laboratory conditions with controlled handling to minimize contamination and external disturbance.

4.3 Test Cases

To verify proper device functionality, each fabricated structure was tested according to its expected electrical behavior.

Test input	Expected output	Observation
Low voltage sweep	Small current flow	Verified
Increasing voltage	Small current flow	Verified
Reverse polarity	Symmetric behaviour expected	Verified

Table 4.1 MSM Device Testing

Test input	Expected output	Observation
Negative bias	High capacitance(accumulation)	Verified
Small positive bias	Reduced capacitance	Verified
Higher positive bias	Transition towards inversion region	Verified

Table 4.2 MOS Capacitor Testing

Test Input	Expected Output	Observation
Gate voltage=0V	Very low drain current	Verified
Increasing gate voltage	Drain current increases	Verified
High drain voltage	Saturation tendency observed	Verified

Table 4.3 TFT Testing

4.4 Safety Measures

During fabrication and testing, standard laboratory safety protocols were followed to ensure safe handling of chemicals, equipment, and electrical instruments.

Chemical Safety

- Gloves and masks were used while handling PANI solution, photoresist, and solvents.
- Chemicals were stored in labeled containers.
- Work was performed in ventilated areas.

Equipment Safety

- High-voltage sputtering systems were operated only under supervision.
- Vacuum chambers were properly sealed before operation.
- Heated surfaces were handled carefully using protective tools.

Cleanliness and Contamination Control

- Substrates were handled using tweezers.
- Dust-free storage was maintained after fabrication.
- Samples were covered when not in use.

Cyber Security / Data Safety

- Experimental data files were stored in password-protected systems.
- Backup copies of measurement results were maintained.
- Unauthorized modification of datasets was avoided.

These measures ensured safe fabrication practices and reliable experimental outcomes.

Chapter 5: Results & Discussion

5.1 Result Analysis

5.1.1 MSM

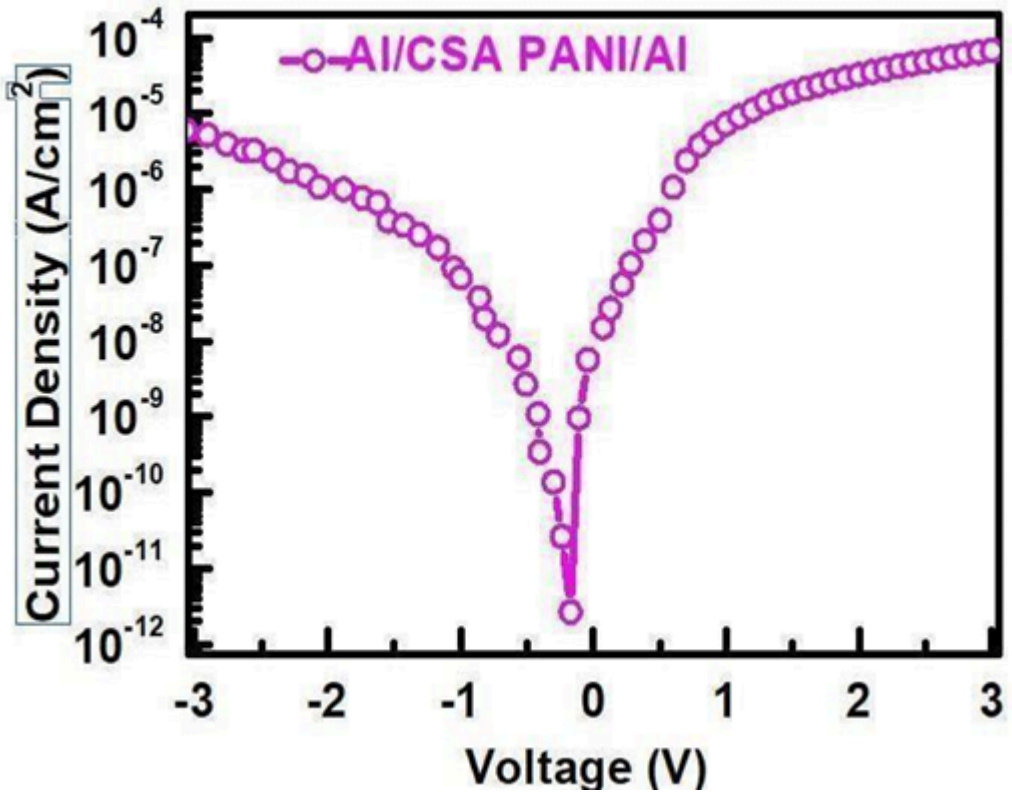


Fig 5.1 I-V Graph OF MSM

The I–V characteristics of a Metal–Semiconductor–Metal (MSM) diode describe how current changes with the applied voltage across its two Schottky contacts. Ideally, an MSM structure would show symmetric current flow for positive and negative voltages because both metal– semiconductor junctions are identical. In reality, the I–V curve shows a low-current region at small voltages due to the Schottky barrier, followed by a rapid increase in current once the applied voltage lowers the effective barrier height. At higher voltages, effects like series resistance and barrier lowering cause the curve to deviate from ideal symmetry. Leakage current is also observed

General C-V Behavior:

Accumulation Region: The capacitance is highest and close to the oxide capacitance (Cox), as a large number of holes accumulate near the oxide- semiconductor interface.

Depletion Region: The capacitance decreases as a depletion region forms and widens with increasing positive gate voltage.

Inversion Region: The capacitance behavior depends on the frequency. At low frequencies, it increases back to Cox. At high frequencies, it saturates at a lower value.

Impact of Temperature: The C-V characteristics of a MOS capacitor are affected by temperature. As temperature increases, the mobility of charge carriers in the semiconductor increases. This affects the rate at which they can respond to changes in the applied voltage. At higher temperatures, the capacitance in the depletion region might decrease as the depletion region widens more easily due to increased carrier mobility Impact of Frequency:

At higher frequencies, the minority carriers might not be able to respond quickly enough to the AC signal used in the measurement. This eventually leads to a decrease in the value of measured capacitance, particularly in the inversion region.

5.1.2 MOS Capacitor

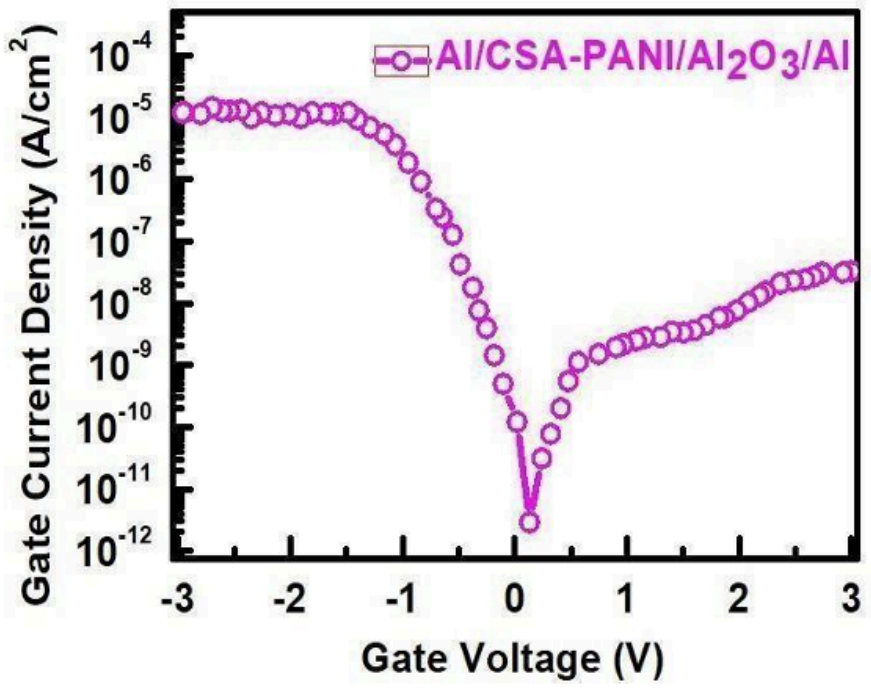


Fig. 5.2: I-V characteristics of Al2O3 dielectric on p-type Si substrate with Al metal contact

I–V Characteristics

The I–V characteristic graph is essential for understanding the electrical behavior and overall performance of the MOS capacitor. It illustrates the relationship between the current (I) flowing through the device and the voltage (V) applied across it.

Ideal MOS Capacitor: In an ideal MOS capacitor with no leakage current, the I–V characteristic would appear as a straight vertical line at V = 0. This represents a purely capacitive device where no current flows regardless of the applied voltage.

Real MOS Capacitor: In practical MOS structures, leakage current is present due to imperfections in the dielectric layer, interface states, and tunneling mechanisms. Therefore, the I–V graph of a real Al_2O_3/PANI-based MOS capacitor exhibits a small but noticeable increase in current as the applied voltage increases. This deviation from ideal behavior provides insight into dielectric quality and leakage pathways.

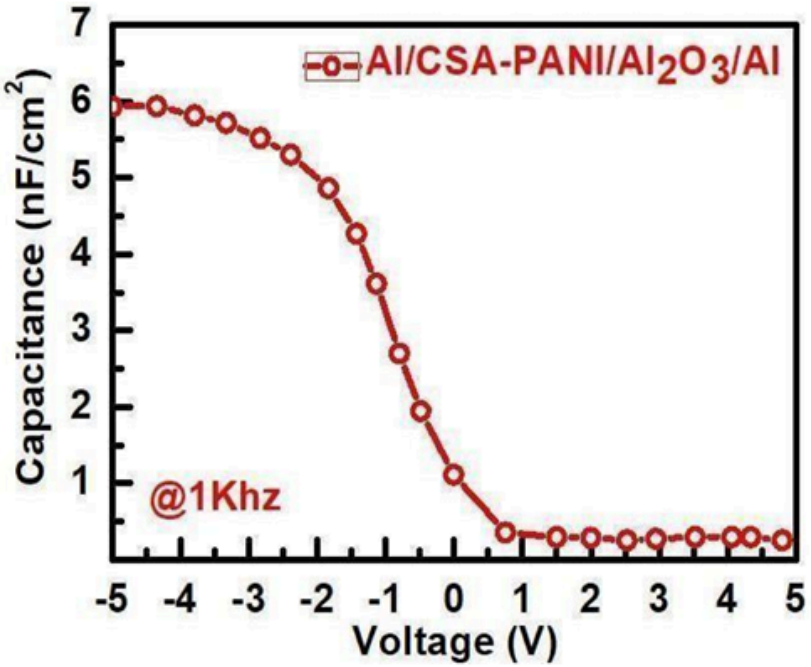


Fig. 5.3: C-V Graph for 1Khz

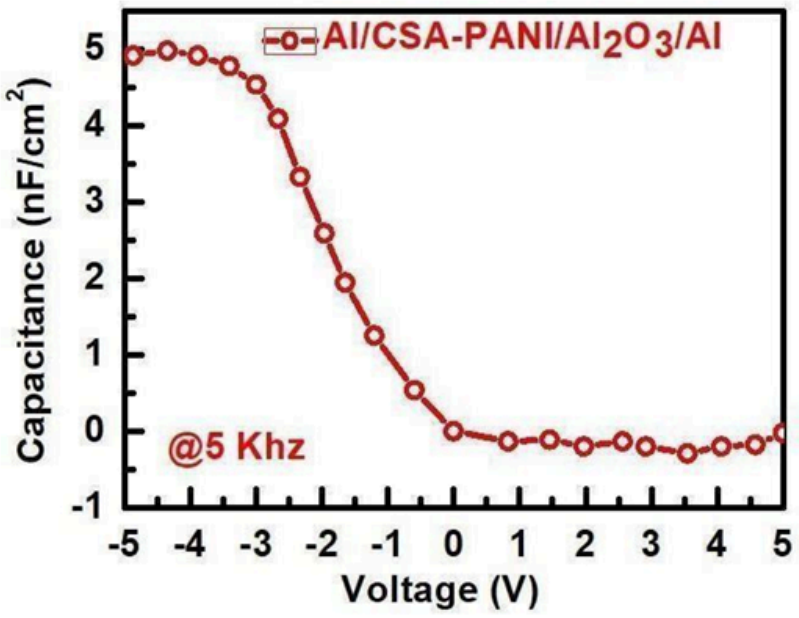


Fig. 5.4: C-V Graph for 5 Khz

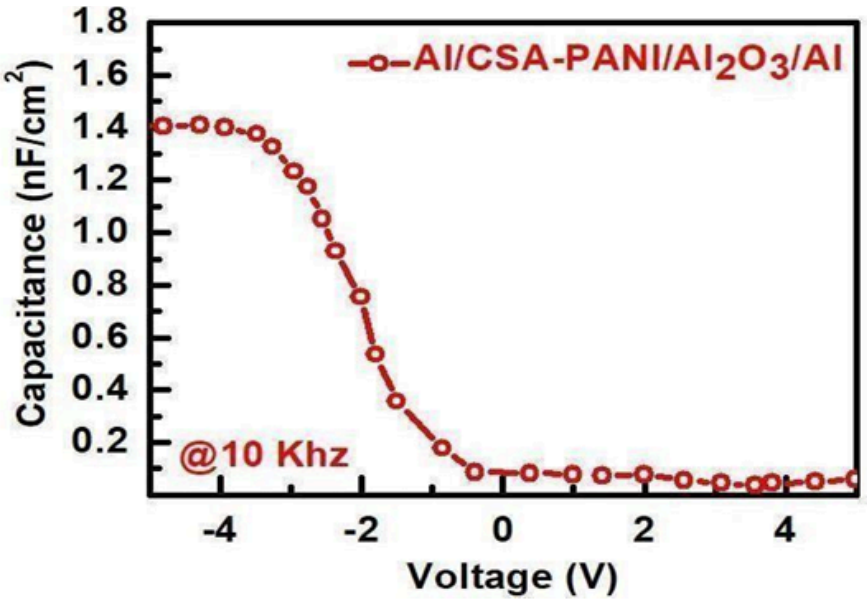


Fig. 5.5: C-V Graph for 10 Khz

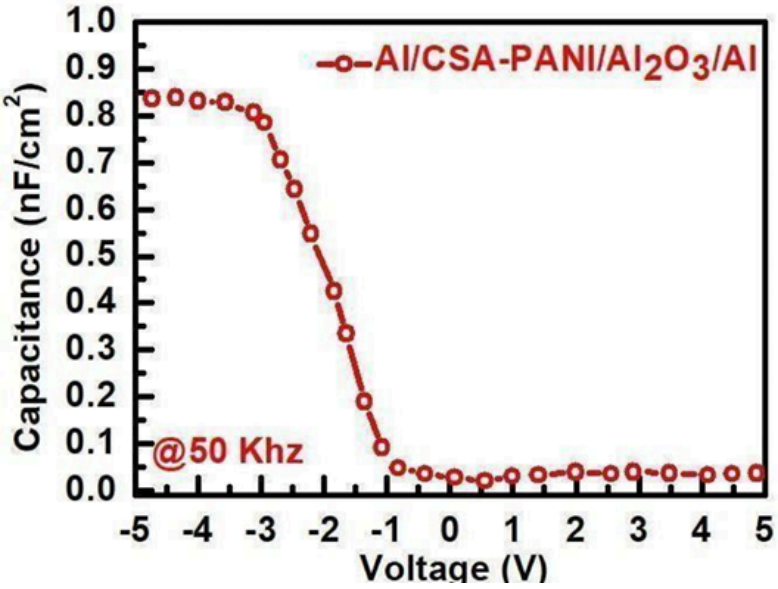


Fig. 5.6: C-V Graph for 50 Khz

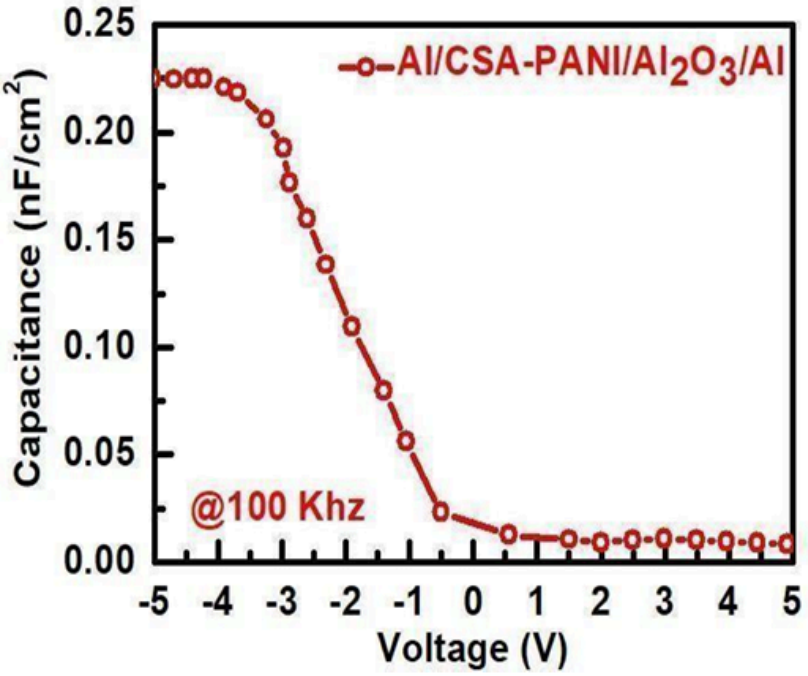


Fig. 5.7: C-V Graph for 100 Khz

PARAMETER	V_{FB}	V_{TH}
C-V Graph for 1 Khz	-2.37	0.53V
C-V Graph for 5 Khz	-2.87	0.14
C-V Graph for 10Khz	-3.12	-0.35
C-V Graph for 50 Khz	-3	-0.7
C-V Graph for 100Khz	-3.4	-0.75

Table 5.1 Comparison of results

C–V Characteristics

The capacitance–voltage (C–V) characteristic is a crucial measurement used to analyze the dielectric behavior and interface properties of an Al_2O_3-based MOS capacitor. It helps determine essential electrical parameters such as oxide thickness, flatband voltage, interface trap density, and doping concentration.

Ideal C–V Behavior: For a perfect dielectric, the C–V curve exhibits sharp transitions among accumulation, depletion, and inversion. Capacitance remains at a maximum in accumulation, decreases smoothly in depletion, and reaches a minimum constant value in inversion. Transitions are abrupt and symmetric.

Real C–V Behavior in Al_2O_3-Based MOS Capacitors: In practical devices, the C–V curve deviates from ideal behavior due to fixed oxide charges, interface traps, and dielectric imperfections. These factors cause stretch-out in the depletion region, shifts in the flatband voltage,

and possible hysteresis between forward and reverse sweeps. High-k dielectrics like Al_2O_3 often exhibit increased inversion capacitance at high frequencies due to charge trapping effects. These deviations provide valuable insight into the dielectric quality and interface condition.

5.1.3 TFT

The fabricated PANI-based Thin Film Transistor (TFT) was electrically characterized using transfer characteristics, output characteristics, and gate leakage current measurements. The obtained results confirm successful transistor action and validate the use of Polyaniline as an active organic semiconductor material.

(a) Transfer Characteristics

The transfer characteristics plot of drain current (Ids) versus gate voltage (Vgs) shows that the drain current increases significantly as negative gate voltage is applied. This indicates p-type conduction behavior of the PANI semiconductor layer.

At gate voltages near 0 V, the drain current remains low, representing the OFF state of the device. As the gate voltage becomes more negative, holes accumulate in the channel region, resulting in enhanced current conduction between source and drain.

The square-root plot of current also shows an approximately linear trend in the operating region, indicating proper field-effect modulation and enabling extraction of threshold voltage.

Key Observations:

- Clear switching behavior observed
- Higher conduction at negative gate bias
- Indicates p-channel TFT operation
- Good gate control over channel current

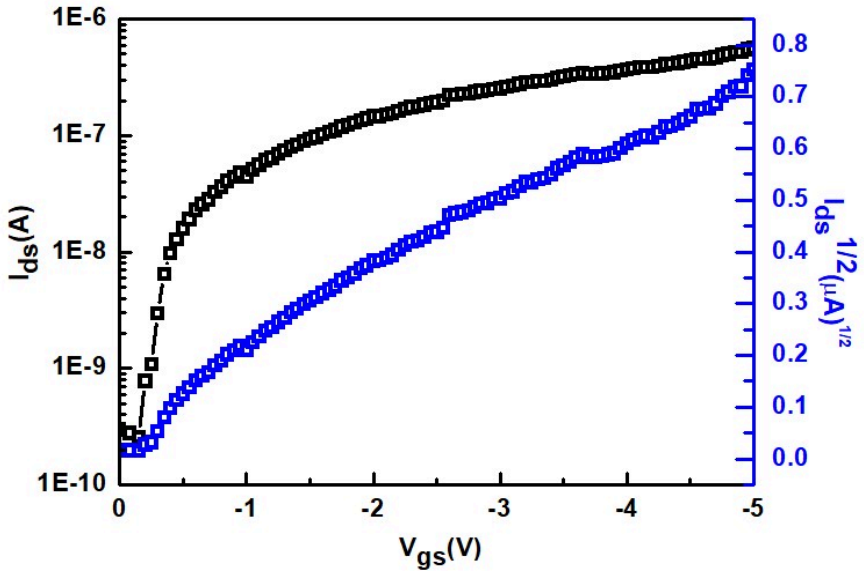


Fig 5.8 Transfer Characteristics of TFT

(b) Output Characteristics

The output characteristics (Ids vs Vds) were measured for different gate voltages ranging from positive to negative bias. It is observed that as the magnitude of negative gate voltage increases, the drain current increases substantially.

For lower drain voltages, the device operates in the linear region where current rises with Vds. At higher drain voltages, the curves gradually approach saturation, which confirms transistor-like channel pinch-off behavior.

Key Observations:

- Drain current increases with increasing negative Vgs
- Linear and saturation regions visible

- Proper modulation of channel conductivity achieved

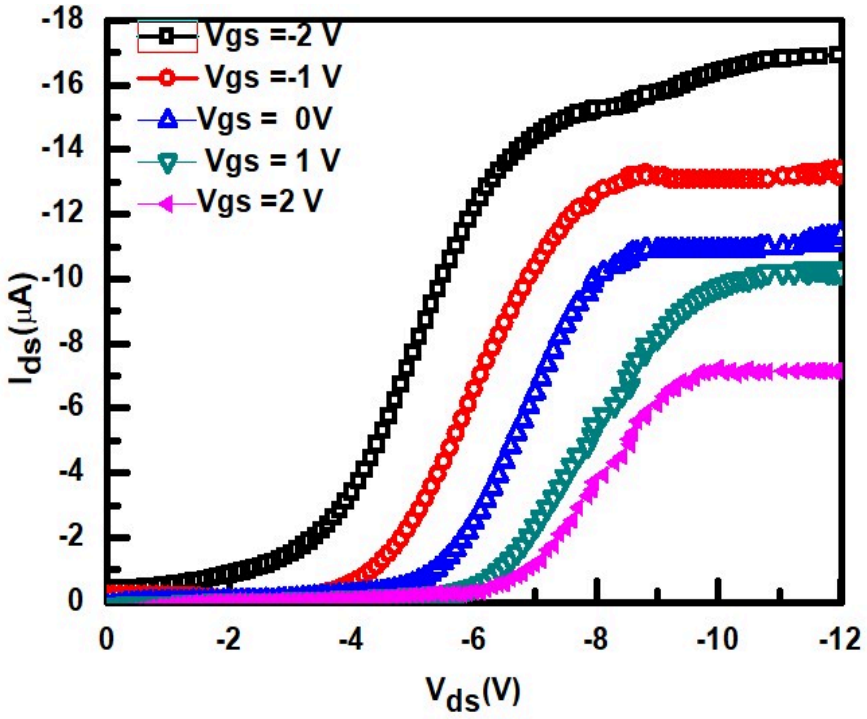


Fig 5.9 Output Characteristics of TFT

(c) Gate Leakage Current

The gate leakage current measurement indicates very low leakage values in the picoampere to nanoampere range near normal operating voltages. This confirms that the deposited Al_2O_3 dielectric layer provides effective insulation between gate and channel.

Low gate leakage is highly desirable because it reduces power loss and improves device reliability.

Key Observations:

- Minimum leakage near zero bias
- Leakage remains low across operating range
- Good dielectric integrity of Al_2O_3 layer

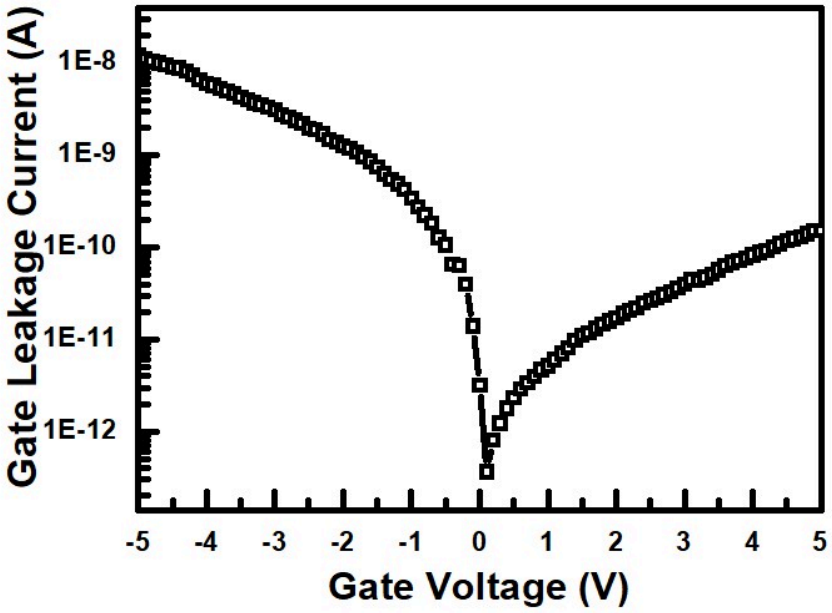


Fig 5.10 Gate Leakage Current of TFT

Result analysis for TFT

Hole Mobility-$\mu \approx \mu_p = 0.02\ cm^2/V \cdot s$

Threshold Voltage -≈ 0.6 V (±0.1)

Gate leakage current (1 V) ~1×10^{-11} A

Semiconductor Type	P-type
Switching Behavior	Achieved
Gate Control	Good
Leakage Current	Low
Saturation Region	Observed
Fabrication Method	Successful

Table 5.2 Performance Summary

5.2 Comparison with Objectives Set in Chapter 1

The project objectives defined in Chapter 1 were successfully addressed through fabrication and testing.

OBJECTIVE	STATUS	REMARKS
Fabricate MSM device using PANI	Achieved	Basic device structure completed
Develop MOS capacitor using Al_2O_3	Achieved	Dielectric behavior verified
Fabricate TFT using PANI	Achieved	Working transistor characteristics obtained

Analyze electrical behavior	Achieved	I-V and leakage responses studied
Optimize fabrication methods	Partially Achieved	Further refinement possible

Table 5.3 Comparison with Objectives

5.3 Economic Analysis

Cost analysis

Compared with conventional silicon device fabrication, the proposed approach offers significant cost reduction because:

- Solution-based processing reduces manufacturing cost
- Lower temperature fabrication saves energy
- Simple laboratory equipment can be used
- Organic materials are comparatively inexpensive
- Suitable for flexible and disposable electronics

Resource Utilization

- Moderate material consumption
- Low semiconductor-grade processing requirement
- Reusable equipment such as spin coater and sputtering system
- Minimal wastage through batch processing

Thus, the project demonstrates economic feasibility for academic research and prototype development.

5.4 Comparison with Existing Solutions and Standards Available

FEATURE	CONVENTIONAL SILICON TFT	ORGANIC PANI TFT

Fabrication Cost	High	Low
Processing Temperature	High	Low
Flexibility	Poor	Good
Mobility	High	Moderate
Large-area Processing	Limited Cost-effectiveness	Good
Mechanical Bendability	Low	High
Complex Manufacturing	High	Low

Compared with standard silicon-based devices, the fabricated PANI TFT offers lower performance in mobility and switching speed, but provides advantages in cost, flexibility, and ease of fabrication.

Compared with previously reported organic devices, the present work successfully demonstrates functional transistor behavior using simple and accessible fabrication techniques, making it suitable for educational laboratories and low-cost electronics research.

The developed devices are promising for applications such as:

- Flexible sensors
- Wearable electronics
- Low-cost switching circuits
- Disposable smart electronics
- Large-area printed electronics

Chapter 6: Conclusion & Future Scope

6.1 Summary of Achievements

This project successfully demonstrated the fabrication and characterization of organic electronic devices using Polyaniline (PANI) as the active semiconductor material. Three important device structures, namely Metal–Semiconductor–Metal (MSM) device, MOS capacitor, and Thin Film Transistor (TFT), were developed using economical and accessible fabrication techniques such as spin coating, photolithography, sputtering, and thermal evaporation.

The fabricated devices were tested through electrical characterization methods including current-voltage and leakage current analysis. The MSM device exhibited expected conduction behavior, the MOS capacitor demonstrated charge storage characteristics, and the TFT showed clear transistor switching action with gate-controlled drain current.

The project also provided practical understanding of thin film deposition, dielectric formation, electrode patterning, interface engineering, and device testing. It confirmed that PANI is a promising material for low-cost and flexible electronics.

Overall, the major achievements of the project are:

- Successful fabrication of three PANI-based device structures
- Demonstration of transistor action in organic TFT
- Low leakage performance using Al_2O_3 dielectric
- Use of simple and cost-effective fabrication methods
- Validation of PANI for organic semiconductor applications
- Practical exposure to semiconductor fabrication processes

6.2 Impact on Society, Environmental Sustainability, Ethical Issues and Compliance

Impact on Society

The development of low-cost organic electronic devices can significantly benefit society by enabling affordable and accessible technology. Devices based on PANI can be used in wearable sensors, health monitoring systems, flexible displays, educational electronics kits, and portable smart systems. Such technologies are especially valuable in cost-sensitive and developing regions.

Environmental Sustainability

Compared with conventional silicon processing, organic electronics offer lower energy consumption due to low-temperature fabrication methods. Solution-based deposition reduces manufacturing complexity and material wastage. Lightweight and flexible devices also reduce resource consumption during packaging and transport.

Use of simple fabrication methods contributes toward greener electronics manufacturing. However, proper disposal of chemical solvents and materials remains important.

Ethical Issues

The project followed ethical engineering practices including honest reporting of results, safe laboratory operation, proper use of institutional resources, and avoidance of data manipulation. No misleading claims regarding commercial performance were made.

Future wearable or sensing applications based on such devices must also ensure user privacy and responsible data handling.

Compliance

All fabrication and testing activities were performed according to standard laboratory safety guidelines. Chemicals were handled carefully, protective equipment was used, and equipment was operated under supervision. Experimental work remained within academic research scope.

6.3 Limitations

Although the project achieved successful device fabrication, certain limitations were observed during development and testing.

- Organic semiconductor performance is lower than silicon devices
- Carrier mobility and switching speed are comparatively limited
- PANI thin film uniformity depends strongly on coating conditions
- Device characteristics may vary due to environmental humidity and temperature
- Advanced cleanroom facilities were not available
- Parameter extraction accuracy was limited by available instruments
- Long-term stability and aging tests were not extensively performed
- Large-scale manufacturing optimization was beyond project scope

These limitations indicate opportunities for further refinement and research.

6.4 Future Work

The present project provides a strong foundation for future development of organic semiconductor devices. Several improvements and extensions can be carried out in subsequent research.

- Improve PANI film quality through optimized synthesis and coating methods
- Use flexible plastic substrates for bendable electronics
- Study long-term stability under humidity and temperature stress
- Explore alternative dielectric materials with higher capacitance
- Increase device miniaturization using advanced lithography
- Improve mobility and ON/OFF ratio through material doping techniques
- Develop printed electronics using inkjet or screen printing methods
- Integrate devices into sensors, wearable systems, and smart tags
- Compare PANI with other organic semiconductors such as PEDOT:PSS or pentacene
- Design complete logic circuits using multiple TFT devices

In future, PANI-based electronics can become an important part of flexible, lightweight, and low-cost next-generation electronic systems.

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PRANATI PANDEY

22102041 POLYANILINE BASED ORGANIC DEVICES

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Sustainability Statement

The present project focuses on the fabrication and analysis of organic electronic devices using Polyaniline (PANI), which offers several environmental and societal advantages compared to conventional semiconductor technologies. The use of organic materials and low-cost fabrication techniques contributes toward sustainable and energy-efficient electronic device development.

One of the major sustainability aspects of this project is the use of Polyaniline, an organic conducting polymer, as the active semiconductor material. Organic semiconductors generally require lower processing temperatures compared to conventional silicon-based technologies, thereby reducing overall energy consumption during fabrication. In this work, thin films were deposited using techniques such as spin coating and sputtering, which are comparatively economical and suitable for scalable manufacturing.

The project also promotes the development of flexible and lightweight electronic devices, which are important for next-generation wearable electronics, sensors, and low-power systems. Organic electronic devices have the potential to reduce manufacturing costs and make electronic technologies more accessible for large-scale societal applications.

From an environmental perspective, the use of controlled thin-film deposition techniques minimizes material wastage. The sequential spin-coating process used for PANI deposition ensures efficient utilization of materials while maintaining film uniformity. Additionally, laboratory safety and waste management practices were followed during chemical handling and fabrication processes to minimize environmental impact.

The fabricated MSM, MOS capacitor, and TFT structures contribute to ongoing research in organic electronics, which aims to develop alternatives to rigid and expensive silicon-based systems. Such technologies may support sustainable electronic manufacturing in the future through low-temperature processing, reduced energy requirements, and compatibility with flexible substrates.

Overall, this project demonstrates the potential of organic semiconductor materials and thin-film fabrication technologies in supporting environmentally conscious and cost-effective electronic device development.

Team Roles and Responsibilities

Both team members actively participated in all stages of the project, including material synthesis, device fabrication, characterization, data analysis, and documentation. Since the project involved multiple fabrication steps requiring close monitoring and coordination, both members were consistently present in the laboratory throughout the work.

🔹 Student A

• Participated in substrate preparation and cleaning processes
• Assisted in photolithography and electrode patterning
• Contributed to sputtering and thermal evaporation processes
• Assisted in preparation and spin coating of PANI thin films
• Performed electrical characterization and graph plotting using OriginPro
• Contributed to report writing, literature survey, and documentation

🔹 Student B

• Participated in synthesis and preparation of Polyaniline (PANI)
• Assisted in thin-film deposition processes including RF and DC sputtering
• Contributed to fabrication of MSM, MOS capacitor, and TFT devices
• Assisted in device characterization and interpretation of I–V and C–V characteristics
• Contributed to data analysis, modelling, and report preparation
• Assisted in compiling experimental observations and final project presentation

🔹 Joint Contributions

• Continuous laboratory monitoring during fabrication processes
• Device optimization and troubleshooting
• Discussion and interpretation of experimental results
• Preparation of figures, flowcharts, and graphs
• Final review and editing of the project report

The project was carried out collaboratively, with both members contributing equally to experimental work and overall project development.